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From Empirical Models to Physics-Informed Neural Networks: The Evolution of Oil Production Forecasting

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ABSTRACT

Production forecasting for oil and gas wells is a decisive element of field-development planning because it directly guides recovery strategy design, production optimisation and risk management. Conventional methods, including empirical decline-curve analysis (DCA) and full-physics numerical simulation, are limited either by their inability to capture complex non-linear flow behaviour or by prohibitive computational requirements. The rise of big data and artificial intelligence has introduced machine learning models such as support vector regression (SVR), random forests (RFs), XGBoost and multi-layer perceptrons (MLPs), whose efficient non-linear fitting has improved predictive accuracy; however, their black-box nature and weak physical consistency now constrain further progress. Modern deep learning (DL) architectures—including long short-term memory (LSTM) or gated recurrent unit (GRU) networks, CNN–LSTM hybrids, transformers, graph convolutional networks (GCNs) and kernel adaptive networks—extend modelling capability to long temporal sequences and systems with multiple interacting wells, fostering a technical shift from purely data-driven learning toward physics-enhanced intelligence. Of particular note, physics-informed neural networks (PINNs) embed Darcy flow equations and related constraints directly in the loss function, which markedly strengthens extrapolation ability and interpretability while offering efficient support for history matching, surrogate modelling and closed-loop reservoir management (CLRM). Nevertheless, these networks still face challenges involving the balance of loss-term weights, multi-scale coupling and training convergence; progress will rely on dynamic weighting schemes and a standardised library of physical priors. This review, therefore, synthesises the evolution from traditional machine learning to physics-constrained approaches in production forecasting, assesses their respective advantages and limitations and identifies future research priorities in high-quality dataset construction, cross-field transfer learning, interpretability enhancement and system-level intelligent optimisation in order to realise fully digital and closed-loop intelligent oilfields.

1 | Introduction

Production forecasting for oil and gas wells is a pivotal component of field-development decision making because it informs recovery planning, guides production optimisation and underpins risk management [1]. Conventional forecasting relies on two principal approaches. The first, often called empirical methods [2, 3], fits decline curves or other statistical relations to historical production data. Although straightforward and widely used in industry, these methods do not incorporate subsurface flow physics and therefore struggle with non-linear dynamics. Model selection is subjective, extrapolation accuracy is limited and different decline models that fit the same data can yield vastly divergent forecasts. Manual removal of outliers and piecewise fitting further complicate standardisation.

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The second approach, numerical reservoir simulation [4], solves governing flow equations to predict production and thus affords a more detailed representation of reservoir conditions. For complex or unconventional reservoirs, however, multi-physics coupling, non-linear flow behaviour and multi-scale effects make model construction difficult. Building a fine geological and engineering model is labour intensive, history matching is time consuming and full-physics simulation of large fields demands substantial computational resources while remaining sensitive to parametric uncertainty. Consequently, real-time forecasting across extensive well networks is challenging if simulation alone is used.

The emergence of big data and artificial-intelligence technologies has introduced machine learning methods as a new avenue for production prediction. Data-driven algorithms automatically extract complex patterns and deliver rapid forecasts, thereby addressing the limitations of traditional techniques in capturing non-linear relationships and in computational efficiency. Many operators are developing digital oilfields and have applied machine learning successfully to seismic interpretation, well-log analysis, pipeline leak detection and drilling optimisation [5]. For production forecasting, numerous studies have trained support vector machines, neural networks and random forests (RFs) on historical well data and have reported encouraging accuracy [6–9].

Nevertheless, purely data-driven models exhibit notable drawbacks. Without physical reasoning they can behave as black boxes, occasionally generating predictions that violate physical laws, such as impossible production or pressure values. Geological and operational conditions vary widely among reservoirs, so a model trained on one dataset often generalises poorly to unseen wells or fields. Sensitivity to noise and outliers further undermines reliability in the absence of physical constraints.To address these issues, researchers have proposed physics-informed neural networks (PINNs), which incorporate governing equations such as Darcy flow into the loss function. PINNs retain high predictive accuracy while improving interpretability, thereby reducing the black box nature of the model. They have shown promise for history matching, surrogate modelling and closed-loop reservoir management (CLRM).

This article reviews the application of machine learning techniques to production forecasting, analyses the current challenges and limitations and outlines future research directions. Emphasis is placed on the integration of physical knowledge with data-driven models, the construction of high-quality datasets, the development of transferable algorithms and the advancement of system-level intelligent optimisation. Some of the references used in this work are tabulated with their scope of work as a summary of their applications in the oil and gas industry as shown in Table 1.

2 | Conventional Machine Learning Methods

Conventional machine learning algorithms include linear regression models [83], distance-based support vector regression (SVR) [84], decision tree ensembles such as RFs [85] and gradient-boosted trees (e.g., XGBoost) [86] and the shallow neural network architecture known as the multi-layer perceptron (MLP) [87]. These methods dispense with explicit physical models and instead infer the input–output mapping directly from data, yielding rapid development and computational efficiency while retaining a strong capacity to approximate complex non-linear relationships.

2.1 | Linear Regression

Linear regression is the most fundamental supervised-learning algorithm in machine learning. Its objective is to establish a linear relationship between a set of features and a target variable by minimising the sum of squared residuals between predicted and observed values—that is, by applying the ordinary least-squares principle. However, when the data exhibit severe multi-collinearity or contain a large number of predictors, the coefficient estimates of ordinary linear regression become unstable, which often leads to overfitting and, consequently, poor generalisation performance. To enhance model robustness and restrain complexity, various regularisation strategies have been proposed.

Lasso regression [88] introduces an L1-norm penalty term into the objective function; the key advantage of this formulation is that it forces many insignificant coefficients to exactly zero. In effect, the model performs automatic feature selection, which is particularly valuable for high-dimensional data analysis and model simplification, while simultaneously improving the interpretability of the results. Ridge regression [89], by contrast, augments the loss function with an L2-norm penalty. This term uniformly shrinks the magnitude of all coefficients without setting any to zero, thereby alleviating the adverse effects of multi-collinearity and stabilising the estimates. Together, these two regularised variants—one emphasising sparsity (L1) and the other emphasising shrinkage (L2)—constitute indispensable tools for linear modelling in complex data environments.

Linear regression and its regularised extensions have been widely applied in petroleum engineering. For instance, Kenzhebek et al. [90] generated a large synthetic dataset with the Buckley–Leverett mathematical model, using reservoir physical properties and water-injection parameters as inputs to predict oil recovery. While ordinary linear regression provided only limited predictive accuracy, the application of an L1 penalty via Lasso markedly improved the goodness of fit and predictive performance [90]. Similarly, Wan et al. [91] analysed extensive datasets comprising reservoir parameters, fracturing designs, treatment variables and production indicators and constructed a ridge-regression model to forecast post-fracturing well productivity. The results demonstrated that ridge regression effectively mitigated the influence of multi-collinearity among variables and produced accurate productivity forecasts [91].

2.2 | SVR

SVR is a supervised-learning model that originated from statistical learning theory and is capable of handling both classification and regression tasks. SVR projects input data into a high-dimensional feature space through a kernel function and then seeks a linear decision or regression boundary in that space. A schematic diagram of its operational principle is presented in Figure 1.

In petroleum engineering, SVR is frequently employed for prediction problems characterised by limited sample size. Bi and Qui [14], for example, developed an intelligent SVR workflow to forecast crude-oil properties, using a hybrid optimisation scheme that

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TABLE 1 | Machine learning models and their applications in oil and gas industry.

Source	Date	Approach	Main area	Model type	Performance indicator	Performance value	Data source	Doi
Khazaeni and Mohaghegh [10]	2011	Prediction	Oil production prediction	KNN	R^2	0.93	Measured data	http://dx.doi.org/10.2118/132643-PA
Golzari et al. [11]	2015	Optimisation	Production strategy optimisation	GA–ANN	Efficiency gain	92%	Numerical model	http://dx.doi.org/10.1016/j.petrol.2015.07.012
Alenezi and Mohaghegh [12]	2016	Prediction	Oil production prediction	ANN	R^2	0.99	Numerical model	http://dx.doi.org/10.1109/KACSTIT.2016.7756063
Anderson [13]	2017	Optimisation	Digital oilfield IoT	PALM integrated IoT platform	Qualitative	—	Measured data	http://dx.doi.org/10.1109/BigData.2017.8268124
Bi and Qiu [14]	2019	Prediction	Oil properties prediction	GA–PSO–SVM	RMSE	0.062	Measured data	http://dx.doi.org/10.1016/j.cjche.2018.12.015
Bikmukhametov and Jäschke [15]	2019	Prediction	Oil production prediction	XGBoost	MAPE	2.02%	Measured data	http://dx.doi.org/10.1016/j.ifacol.2019.06.114
Wang et al. [16]	2019	Prediction	Oil production prediction	DNN	R^2	0.9	Measured data	http://dx.doi.org/10.1016/j.petrol.2019.04.016
Khan et al. [17]	2019	Prediction	Oil rate prediction	ANN	R^2	0.99	Measured data	http://dx.doi.org/10.2118/194713-MS
Guevara et al. [18]	2019	Prediction	Oil production prediction	GAM, SCAM	Qualitative	—	Measured data	http://dx.doi.org/10.2118/195690-PA
Fath et al. [19]	2020	Prediction	SGOR prediction	RBF-ANN	R^2	0.978	Measured data	http://dx.doi.org/10.1016/j.petlm.2018.12.002
Yang et al. [20]	2020	Prediction	Bubble point pressure prediction	ANN	MARE	5.15%	Measured data	http://dx.doi.org/10.1016/j.petrol.2019.106598
Song et al. [21]	2020	Prediction	Oil production prediction	LSTM	MAPE	9.52%–9.88%	Measured data	http://dx.doi.org/10.1016/j.petrol.2019.106682
Liu et al. [22]	2020	Prediction	Oil production prediction	LSTM	MAPE	2.73%	Measured data	http://dx.doi.org/10.1016/j.petrol.2020.107013
Kim et al. [23]	2020	Optimisation	Well placement optimisation	PSO–CNN	R^2	0.93	Numerical model	http://dx.doi.org/10.1016/j.petrol.2020.107424
Morozov et al. [24]	2020	Prediction	Oil production prediction	CatBoost	R^2	0.815	Measured data	http://dx.doi.org/10.1016/j.petrol.2020.107504
Zhong et al. [25]	2020	Prediction	Oil production prediction	cDC–GAN	R^2	>0.977	Numerical model	http://dx.doi.org/10.1016/j.petrol.2020.107574
Wang et al. [26]	2020	Prediction	Oil production prediction	LSTM	R^2	0.93	Measured data	http://dx.doi.org/10.1016/S1876-3804(20)60119-7
Bao et al. [27]	2020	Prediction	Oil production prediction	EnKF–LSTM	R^2	0.96	Measured data	http://dx.doi.org/10.2118/199005-MS
Bagheri et al. [28]	2020	Prediction	Oil production prediction	LSTM	R^2	0.98	Measured data	http://dx.doi.org/10.4043/30854-MS
Parhi and Patwardhan [29]	2021	Prediction	Oil production prediction	ANN	Qualitative	—	Measured data	http://dx.doi.org/10.1007/978-981-16-0401-0_36
Obite et al. [30]	2021	Prediction	Oil production prediction	ARIMA, ANN, RF	Qualitative	—	Measured data	http://dx.doi.org/10.1016/j.egyr.2021.04.050

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TABLE 1 | (Continued)

Source	Date	Approach	Main area	Model type	Performance indicator	Performance value	Data source	Doi
Fan et al. [31]	2021	Prediction	Oil production prediction	LSTM	MAPE	14.40%	Measured data	http://dx.doi.org/10.1016/j.energy.2020.119708
Pan et al. [32]	2021	Prediction	Oil production prediction	Attention–CNN–LSTM	RMSE	0.315–0.402	Measured data	http://dx.doi.org/10.1088/1742-6596/2030/1/012038
He et al. [33]	2021	Prediction	Gas production prediction	RF	R^2	0.928	Measured data	http://dx.doi.org/10.13639/j.odpt.2021.04.518
Hardanto [34]	2021	Optimisation	EOR optimisation	MRGC + DNNA	MCC	0.85	Measured data	http://dx.doi.org/10.29118/IPA21-G-6
Han and Kwon [35]	2021	Prediction	Gas production prediction	PINN	MAPE	22.18%	Measured data	http://dx.doi.org/10.3390/en14123629
Hung et al. [36]	2022	Prediction	Oil production prediction	LSTM, GRU, RNN	Qualitative	—	Measured data	http://dx.doi.org/10.1007/978-981-16-7735-9_50
Wevill et al. [37]	2022	Prediction	Well classification	RF	Accuracy	97%	Measured data	http://dx.doi.org/10.1016/B978-0-12-822295-9.00007-8
Zha et al. [38]	2022	Prediction	Gas production prediction	CNN–LSTM	MAPE	7.70%	Measured data	http://dx.doi.org/10.1016/j.energy.2022.124889
Almajid and Abu-Al-Saud [39]	2022	Prediction	Oil production prediction	PINN	Qualitative	—	Numerical model	http://dx.doi.org/10.1016/j.petrol.2021.109205
Chen et al. [40]	2022	Prediction	EUR prediction	*K*-means + DCA	R^2	0.14–0.28	Measured data	http://dx.doi.org/10.1016/j.petrol.2021.109216
Li et al. [41]	2022	Prediction	Oil production prediction	SSA–BiGRU	R^2	0.89–0.95	Measured data	http://dx.doi.org/10.1016/j.petrol.2021.109309
Rodriguez and Salazar [42]	2022	Prediction	Oil production prediction	LSTM	RMSE	<226	Numerical model	http://dx.doi.org/10.1016/j.petrol.2021.109715
Li et al. [43]	2022	Prediction	Oil production prediction	PSO–CNN–LSTM	R^2	0.93	Measured data	http://dx.doi.org/10.1016/j.petrol.2022.110586
Li et al. [44]	2022	Prediction	Gas production prediction	TCN	MAPE	<6%	Measured data	http://dx.doi.org/10.1016/j.petrol.2022.110644
Lu et al. [45]	2022	Optimisation	Fracturing parameter optimisation	PSO–DNN	NPV increase	USD 680,000	Numerical model	http://dx.doi.org/10.1016/j.petrol.2022.110900
Zhen et al. [46]	2022	Prediction	Gas production prediction	TCN	MAPE	<5.9%	Measured data	http://dx.doi.org/10.1016/j.petrol.2022.111043
Zhang et al. [47]	2022	Prediction	Oil production prediction	TCN	MAPE	4.96%	Measured data	http://dx.doi.org/10.1016/S1876-3804(22)60339-2
Zhai et al. [48]	2022	Prediction	EUR prediction	SVR	R^2	0.84	Measured data	http://dx.doi.org/10.1080/15567036.2022.2100521
Parhi and Patwardhan [49]	2022	Prediction	Parameter estimation for DCA	ANN	Qualitative	—	Measured data	http://dx.doi.org/10.1149/10701.10701ecst
Rahmanifard et al. [50]	2022	Prediction	Oil production prediction	MLP	MSE	0.00005	Numerical model	http://dx.doi.org/10.15530/urtec-2022-3703284

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TABLE 1 | (Continued)

Source	Date	Approach	Main area	Model type	Performance indicator	Performance value	Data source	Doi
Du et al. [51]	2022	Prediction	Oil production prediction	GCN–GRU	MAPE	2.10%	Measured data	http://dx.doi.org/10.2118/210575-PA
Zhu et al. [52]	2022	Prediction	Oil production prediction	LSTM	SSE	0.0005	Numerical model	http://dx.doi.org/10.32604/fdmp.2022.020942
Vikara and Khanna [53]	2022	Prediction	Multi-phase production prediction	LSTM	R^2	0.91	Measured data	http://dx.doi.org/10.3390/pr10040740
Ibrahim et al. [54]	2022	Prediction	Multi-phase production prediction	ANN, RNN, XGBoost	R^2	0.96–0.98	Measured data	http://dx.doi.org/10.3390/s22145326
Li et al. [55]	2022	Prediction	Multi-phase production prediction	BiGRU	R^2	>0.73	Measured data	http://dx.doi.org/10.2118/212290-PA
Zhou et al. [56]	2023	Prediction	Oil production prediction	CNN–BiGRU–AM	R^2	>0.89	Measured data	http://dx.doi.org/10.1016/j.apenergy.2023.121249
Zhao et al. [57]	2023	Prediction	Gas production prediction	GCN	R^2	>0.9	Measured data	http://dx.doi.org/10.1016/j.cageo.2023.105312
Kim et al. [58]	2023	Prediction	Gas production prediction	RF + GB	R^2	0.819	Measured data	http://dx.doi.org/10.1016/j.cageo.2023.105452
Pan et al. [59]	2023	Prediction	Oil production prediction	CNN–LSTM–SA	MAPE	13.70%	Measured data	http://dx.doi.org/10.1016/j.energy.2023.128701
Du et al. [60]	2023	Prediction	Oil production prediction	GCN–LSTM	RMSE	0.72 t/d	Measured data	http://dx.doi.org/10.1016/j.geoen.2023.211542
Dong et al. [61]	2023	Prediction	Oil production prediction	XGBoost	MAPE	5.87%	Measured data	http://dx.doi.org/10.1016/j.geoen.2023.212402
Song et al. [62]	2023	Prediction	Oil production prediction	Light GBM	R^2	0.6	Measured data	http://dx.doi.org/10.1088/1742-6596/2468/1/012084
Osah and Howell [63]	2023	Prediction	Oil production prediction	SVR	R^2	0.65–0.76	Measured data	http://dx.doi.org/10.1144/petgeo2022-071
Mousavi et al. [64]	2023	Optimisation	Well placement optimisation	XGBoost, light GBM, CatBoost	R^2	>0.95	Numerical model	http://dx.doi.org/10.2118/217467-PA
Song et al. [65]	2023	Prediction	Oil production prediction	CNN–LSTM	R^2	0.95	Measured data	http://dx.doi.org/10.3390/en16062904
Jiang et al. [66]	2023	Prediction	Oil production prediction	PINN	MSE	<0.001	Numerical model	http://dx.doi.org/10.3390/w15020218
Lei [67]	2023	Prediction	Reserve prediction	RF	R^2	0.88	Measured data	http://dx.doi.org/10.3969/j.issn.1000-105X.2023.03.014
Lei et al. [68]	2023	Optimisation	Fracturing parameter optimisation	PSO–RF	Qualitative	—	Measured data	http://dx.doi.org/10.3969/j.issn.2095-1132.2023.02.008
Al-Obaidi et al. [69]	2023	Prediction	Reservoir pressure prediction	RF	MAPE	6.80%	Measured data	http://dx.doi.org/10.53656/nat2023-3-4.01
Fargalla et al. [6]	2024	Prediction	Gas production prediction	CNN–BiGRU	R^2	0.97	Measured data	http://dx.doi.org/10.1016/j.energy.2023.130184

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TABLE 1 | (Continued)

Source	Date	Approach	Main area	Model type	Performance indicator	Performance value	Data source	Doi
Zhou et al. [70]	2024	Prediction	Oil production prediction	Duong–CNN–LSTM	R^2	0.91–0.96	Measured data	http://dx.doi.org/10.1016/j.engeos.2023.100252
Chen et al. [71]	2024	Prediction	Oil production prediction	CNN–GRU	MAPE	<4.7%	Measured data	http://dx.doi.org/10.1016/j.geoen.2023.212528
Xu and Leung [72]	2024	Prediction	Gas production prediction	GRU	RMSE	0.22MCF/d	Measured data	http://dx.doi.org/10.2118/215056-PA
Du et al. [73]	2024	Prediction	Oil production prediction	GRU	R^2	0.99	Measured data	http://dx.doi.org/10.1016/j.energy.2023.129688
Wang et al. [74]	2024	Prediction	Gas production prediction	PINN	R^2	>0.81	Measured data	http://dx.doi.org/10.1016/j.eswa.2024.123387
Al Dwood et al. [75]	2024	Prediction	Oil production prediction	ANN–GB–LR	R^2	0.998	Measured data	http://dx.doi.org/10.1016/j.flowmeasinst.2024.102690
Huang et al. [76]	2024	Prediction	Oil production prediction	GNN	Accuracy	91.70%	Measured data	http://dx.doi.org/10.1016/j.petsci.2023.11.008
Gu [77]	2024	Prediction	Fracturing performance evaluation	SVR	MAPE	15.67%	Measured data	http://dx.doi.org/10.1088/1742-6596/2834/1/012095
Cui et al. [78]	2024	Prediction	Oil production prediction	LSTM	R^2	0.82	Measured data	http://dx.doi.org/10.1177/01445987231188161
Liu et al. [79]	2024	Optimisation	Completion sequence optimisation	RF, GBDT, LSTM	RMSE	7.35–20.01	Measured data	http://dx.doi.org/10.48550/arXiv.2402.15608
Motaei et al. [80]	2024	Prediction	Oil production prediction	Linear regression	R^2	0.99	Measured data	http://dx.doi.org/10.51865/jpgt.2024.02.22
Bai et al. [81]	2024	Prediction	Oil production prediction	Light GBM	MAPE	6.31%	Measured data	http://dx.doi.org/10.1016/j.petsci.2024.02.018
Gao et al. [82]	2025	Prediction	Oil production prediction	ANN	RMSE	0.051	Measured data	http://dx.doi.org/10.1038/s41598-025-94294-6

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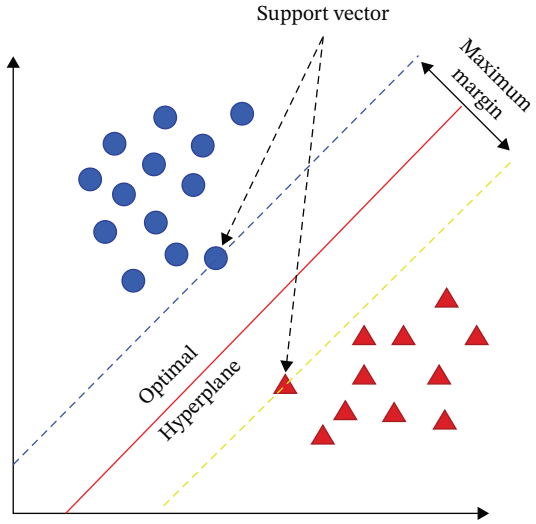

FIGURE 1 | Support vector regression.

combines genetic algorithms (GAs) with particle-swarm optimisation (PSO) to fine-tune the SVR hyperparameters. This approach substantially enhanced model accuracy [14]. Ng et al. [92] likewise coupled SVR with a meta-heuristic algorithm to optimise hyperparameters and achieved high-precision forecasts on well-production data from the Volve field; in their experiments, SVR delivered superior accuracy and exhibited excellent adaptability to small-sample conditions.

The strengths of SVR include strong generalisation capability for small or high-dimensional datasets and an effective treatment of non-linear relationships through appropriate kernel choices. Nevertheless, SVR performance is sensitive to the selection of kernel functions and hyperparameters, and its training phase can be computationally intensive. Addressing these challenges, Shevade et al. [93] refined the SVR training algorithm, improving convergence speed and reducing memory requirements, thereby expanding the applicability of SVR to large-scale datasets.

2.3 | Decision Tree Ensemble Methods

2.3.1 | RFs

Decision tree ensemble approaches are widely valued for their proficiency in capturing non-linear relationships while simultaneously performing implicit feature selection. RFs, proposed by Breiman [94], constitute a quintessential bagging-based ensemble technique. As depicted in Figure 2, the algorithm assembles a large collection of decision tree base learners through a bagging strategy: multiple training subsets are sampled with replacement from the original dataset, and a separate decision tree is trained on each subset.

The principal innovation of RF is the introduction of feature randomness. At every node split in every tree, only a randomly chosen subset of features—rather than the full set—is considered. The ensemble's final prediction is generated by averaging the outputs of all constituent trees. This dual perturbation of data and features markedly reduces variance, thereby bolstering both generalisation capability and robustness.

Because RF effectively quantifies feature importance and accommodates complex non-linear interactions, it has become a popular tool for production forecasting and factor-impact analysis in petroleum engineering. For instance, Xue et al. [95] developed a multi-objective RF regression framework that simultaneously addresses several production metrics when forecasting shale-gas well output, outperforming single-objective models. Wevill et al. [37] compared RF, support vector machines and a single decision tree in classifying production levels of unconventional shale wells; RF achieved the highest accuracy, support vector machines ranked second and the single-tree model lagged far behind, underscoring the superior pattern-recognition capability of ensemble learning methods in intricate reservoir contexts.

2.3.2 | Gradient-Boosted Decision Trees (GBDTs)

GBDTs, exemplified by XGBoost, constitute a boosting-based ensemble algorithm [96]. As illustrated in Figure 3, GBDT

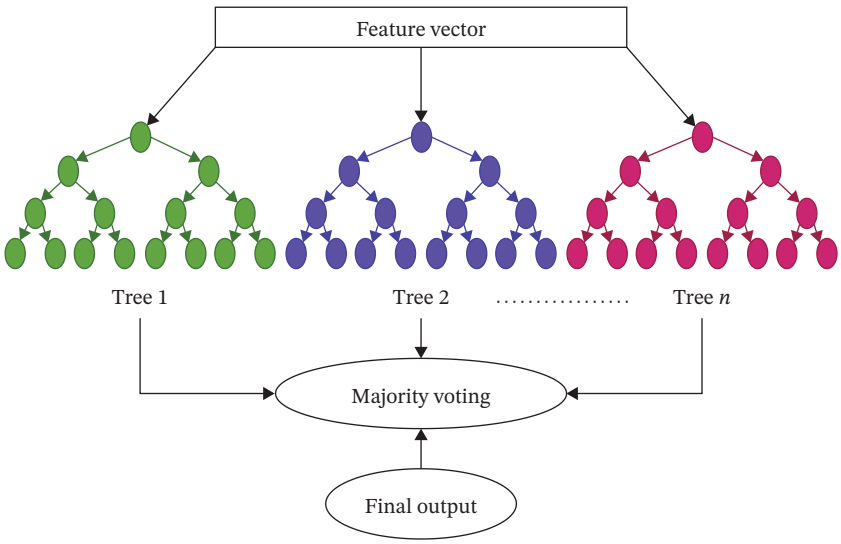

FIGURE 2 | Random forests.

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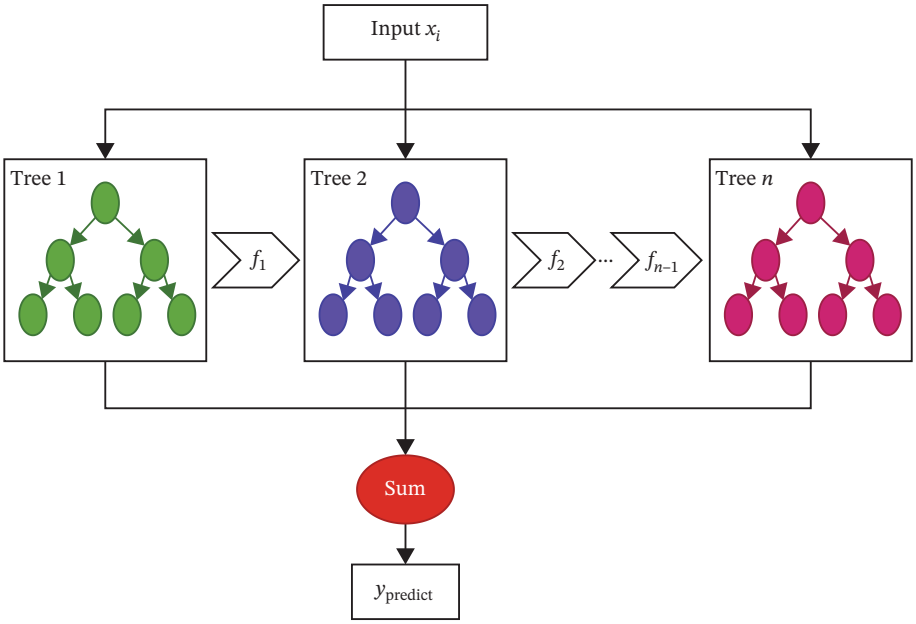

FIGURE 3 | Gradient-boosted decision trees.

incrementally improves predictive accuracy by sequentially fitting the residuals of prior models. A series of decision tree weak learners are trained in turn, and each new tree is constructed to correct the prediction error left by the preceding ensemble. Leveraging gradient-boosting theory, the algorithm employs the first- and second-order derivatives of the loss function to guide optimisation, thereby effectively minimising the objective loss.

XGBoost introduces several significant enhancements over the conventional GBDT framework. Most notably, it incorporates explicit regularisation terms into the objective function, which provide direct control over model complexity and thus mitigate the risk of overfitting. In a shale-oil study, Liu et al. [97] developed an XGBoost model to forecast the productivity of hydraulically fractured horizontal wells and coupled it with Shapley additive explanations (SHAPs) values to elucidate feature contributions. The resulting model not only delivered accurate post-fracturing production forecasts but also quantitatively clarified the influences of reservoir and treatment parameters, thereby enhancing transparency and credibility [97]. Dong et al. [61] applied XGBoost to predict the early-time productivity of oil wells and achieved an accuracy exceeding 94 %.

Decision tree-based models excel at handling features of varied scales and are readily parallelisable; however, they remain fundamentally interpolation tools—predictive reliability diminishes outside the range of training data, and their extrapolation capability is limited, particularly for extreme operating conditions. These shortcomings can be partially alleviated through ensemble fusion with other algorithmic families or through appropriate regularisation strategies.

2.4 | Artificial Neural Networks (ANNs)

ANNs constitute another major family of machine learning methods, among which the MLP is the most fundamental and widely adopted architecture [98]. As illustrated in Figure 4, an MLP consists of an input layer, one or more non-linear hidden layers and an output layer, each comprising multiple neurones. Signals

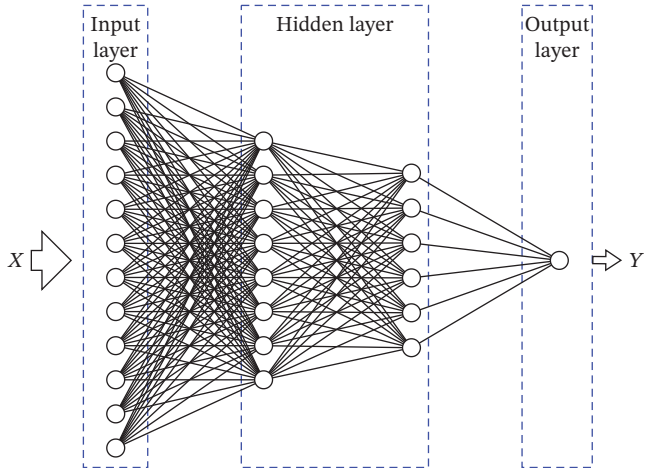

FIGURE 4 | Multi-layer perceptron (MLP).

propagate forward through weighted connections; every neurone computes a weighted sum of its inputs and applies a non-linear activation function, thereby enabling the network to approximate highly complex and non-linear input–output mappings.

Training proceeds by means of the back-propagation algorithm combined with a gradient-descent optimiser. The loss function quantifies the discrepancy between the network's predictions and the true targets; its gradient is back-propagated layer by layer, iteratively updating the weights and biases to minimise the loss. Owing to their universal approximation property, MLPs have been employed to address a variety of reservoir-engineering problems.

Prasetyo combined empirical mode decomposition (EMD) with a high-order MLP to improve production-forecast accuracy for wells in India's Cambay Basin and Indonesia's Sumatra Basin [99]. Elmabrouk et al. [100] developed a multi-layer MLP model that uses readily available operational parameters—such as daily production history, completion depth and wellbore diameter—to

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forecast the future output of newly drilled wells without recourse to detailed geological or flow-simulation data. Tested on a Libyan oilfield, the model exhibited strong generalisation and high predictive accuracy, demonstrating practicality for early-stage capacity assessment and development planning [100]. Aizenberg et al. [101] applied a multi-layer MLP to production forecasting and, using real-field datasets, showed that the network captures reservoir production trends effectively, even under complex subsurface dynamics. Cao et al. [102] integrated geological maps, historical production, pressure data and operational constraints into an ANN to predict well productivity; the inclusion of diverse data sources significantly boosted forecast precision and enabled the model to accommodate the challenging geological and operational conditions characteristic of unconventional resource plays, thereby offering a reliable decision-support tool for production optimisation.

The chief strength of ANNs lies in their powerful generalisation ability: with a sufficient number of hidden nodes, they can approximate any continuous function. Their limitations include a propensity for overfitting, a large number of parameters and a substantial demand for training data and appropriate regularisation.

In summary, conventional machine learning methods provide a diverse suite of modelling strategies for forecasting oil and gas production. Linear regression and its regularised extensions combine transparency with parsimony and perform reliably when relationships among variables remain largely linear. Support vector machines excel in small-sample and high-dimensional contexts. Ensemble techniques such as RFs and XGBoost bolster robustness and predictive accuracy by aggregating multiple weak learners. ANNs, exemplified by MLPs, leverage strong non-linear-mapping capacity and are particularly advantageous when integrating multi-source data and modelling complex dynamic behaviour. Each approach strikes a different balance among accuracy, generalisation capability, training complexity and interpretability, so practical deployment requires a judicious selection or combination tailored to specific application objectives and data characteristics. Table 2 summarises the principal strengths, limitations and recommended application scopes of these conventional machine learning methods.

3 | Deep Learning (DL) Methods

DL, an outgrowth of conventional machine learning, employs multi-layer neural networks and is renowned for its exceptional capacity to fit and represent large-scale data sets [103]. Oil- and gas-production forecasting constitutes a prominent branch of time-series prediction, and DL is particularly well suited to the high-dimensional and long-horizon nature of such problems. Unlike shallow models that rely on a fixed historical window, DL architectures equipped with recurrent or attention mechanisms can capture long-range dependencies far exceeding the window length. In reservoir engineering, DL has progressed from exploratory trials to several specialised directions, including long-term well-production forecasting, injector-producer connectivity analysis, anomaly detection and reconstruction in production data and image-recognition-assisted prediction. The following subsections introduce representative DL models and their applications in production forecasting (Sections 3.1–3.4).

3.1 | Recurrent Neural Network (RNN) Models

RNNs are expressly designed for sequential data modelling [104]. Their defining feature is the propagation of hidden states through recurrent connections at successive time steps, allowing the network to maintain a dynamic memory of historical information. Vanilla RNNs, however, struggle with gradient vanishing and explosion when learning long-range dependencies. To address this limitation, gated variants—most notably the long short-term memory (LSTM) and the gated recurrent unit (GRU)—were developed; their selective-memory mechanisms markedly improve the modelling of long sequences.

An LSTM comprises an input layer, one or more hidden layers arranged as a chain of recurrent cells, and an output layer. Each cell contains three gates—a forget gate, an input gate and an output gate—that regulate information flow. The forget gate discards irrelevant components of the previous cell state, the input gate stores new information in the current cell state and the output gate transmits pertinent information to the next state [105]. Gate operations employ activation functions such as sigmoid and tanh: the sigmoid function modulates information flow, whereas tanh constrains hidden-state values within$(-1, 1)$. The final hidden state is passed to the output layer for prediction.

The GRU is a streamlined variant of the LSTM that implements two principal modifications: the three gates are reduced to an update gate and a reset gate, and the cell state is merged with the hidden output into a single vector [106]. With fewer parameters than the LSTM, the GRU generally achieves comparable performance at lower computational cost. The architectures of the vanilla RNN, LSTM and GRU are illustrated in Figure 5.

RNNs and their gated variants—LSTM networks and GRUs—model sequential dependence by recursively updating hidden states, making them well suited to time-series signals such as well production rates and water cut. Misra et al. [107] utilised actual production and operational data from the Volve field in the Norwegian North Sea to train an LSTM, achieving encouraging accuracy and underscoring the model's potential for production forecasting. He proposed an LSTM augmented with a self-attention mechanism (SA-LSTM) for tight-oil wells; the hybrid network exhibited markedly superior performance in capturing both long- and short-term dependencies and demonstrated enhanced robustness to noise [108]. Bao et al. [27] further integrated ensemble Kalman filtering (EnKF) with an LSTM to permit real-time parameter correction, thereby creating an 'enhanced LSTM' for reservoir-production forecasting. Under a five-spot water flood pattern, the EnKF–LSTM continuously updated its internal weights as new production data arrived, driving the predicted oil-rate and water-cut curves into closer agreement with field observations. This concept illustrates the strength of coupling DL with data assimilation: the model is not static after initial training but continues to 'learn' from incoming data, achieving a semi-physics and semi-data-driven closed-loop forecast [27].

To address the inability of conventional methods (e.g., ARIMA or shallow neural networks) to capture high-dimensional, non-linear, long-term dependencies—and the high computational cost of LSTMs—Al-Shabandar et al. [109] devised a deep GRU (DGRU) network comprising multiple stacked GRU layers with dropout regularisation. Using 90 years of U.S. oil-production data, the DGRU substantially outperformed RNNs and LSTMs in

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TABLE 2 | Summary of strengths and limitations of conventional machine learning methods.

Method	Principal strengths	Key limitations	Typical application scope
Linear regression (ordinary, ridge, Lasso)	• Simple, fast to train • Coefficients are directly interpretable • Regularisation (ridge/Lasso) mitigates multi-collinearity and overfitting	• Captures only linear relationships • Sensitive to outliers • Performance deteriorates when true dynamics are strongly non-linear	Quick screening or baseline models; reservoirs whose production trends are approximately linear in selected variables; studies requiring transparent coefficient analysis
Support vector regression (SVR)	• Strong generalisation on small or high-dimensional datasets • Kernel trick handles moderate non-linearity • Robust to outliers near the decision surface	• Choice of kernel and hyperparameters strongly affects accuracy • Training cost scales poorly with sample size • Limited interpretability	Wells with limited historical data; early-life production prediction; cases where high-dimensional geologic or operational descriptors are available but samples are few
Random forest (RF)	• Handles complex non-linear interactions • In-built assessment of feature importance • Resistant to overfitting and outliers • Easy to parallelise	• Predictions are ensemble averages (loss of fine detail) • Model size can become large • Limited extrapolation beyond training range	Factor-impact studies; production classification or ranking; quick-response forecasting tools for routine field surveillance
Gradient-boosted decision trees (GBDT/XGBoost)	• High predictive accuracy via sequential residual fitting • Handles mixed data types and missing values • Built-in regularisation and early stopping control complexity • Efficient, scalable implementation	• Hyperparameter tuning can be complex • Less interpretable than RF unless explainability tools (e.g., SHAP) are applied • Still fundamentally interpolative	Short- and medium-term production forecasting where accuracy is paramount; sensitivity analyses using SHAP/feature importance; decision support under tight computational budgets
Multi-layer perceptron (MLP) artificial neural network	• Universal approximator capable of capturing highly non-linear mappings • Flexible architecture accommodates multi-source, time-varying inputs • Once trained, inference is very fast	• Large number of parameters, prone to overfitting • Requires substantial, well-curated data • Opaque internal representations without additional explainability methods	Integration of heterogeneous data streams (logs, pressures and operations); reservoirs exhibiting complex, coupled dynamical behaviour

Note: Hybrid or stacked models can combine complementary strengths (e.g., SVR residuals refined by GBDT or MLP ensembles regularised with dropout and batch normalisation) for further performance gains.

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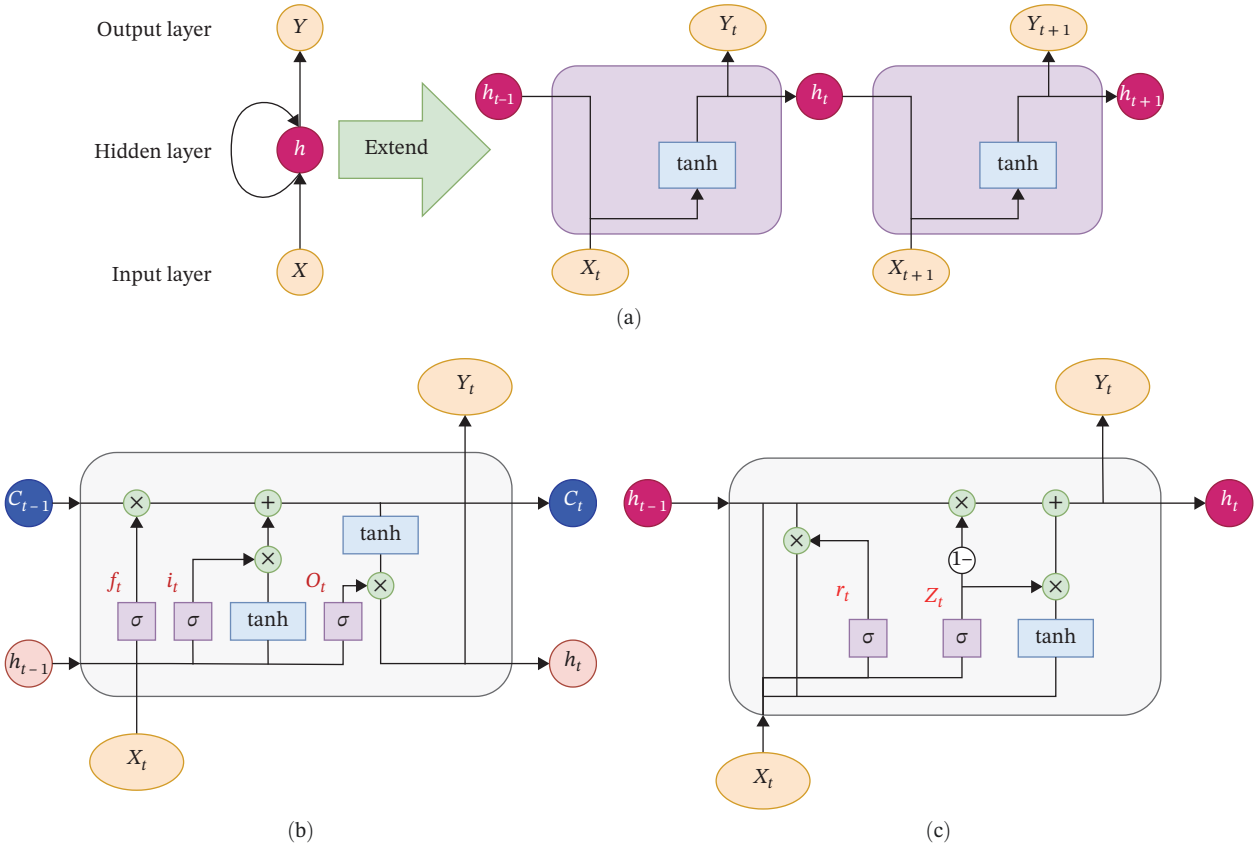

FIGURE 5 | Schematic diagram of three recurrent neural network architectures. (a) Schematic diagram of the RNN architecture: X_t denotes the model input at time t, H_t the hidden state at time step t, Y_t the model output at time t, and tanh represents the hyperbolic-tangent activation function. (b) Schematic diagram of the LSTM architecture: f_t is the forget gate, which determines which components of the current input X_t and the previous hidden state h_{t-1} are unimportant and should be discarded; i_t is the input gate, responsible for extracting salient information from the input; O_t is the output gate, which decides the information to emit at time step t; C_t denotes the cell state at time t; σ represents the sigmoid activation function. (c) Schematic diagram of the GRU architecture: r_t is the update gate, which decides which portions of the incoming information are to be forgotten or retained, whereas Z_t is the reset gate, which controls the extent of forgetting, its value ranges from 0 to 1, with values closer to 0 indicating a higher degree of forgetting.

predictive accuracy while reducing training time to roughly one-tenth that of an LSTM. The GRU's gating mechanism mitigated vanishing-gradient issues and effectively tracked long-interval dynamics, providing an innovative and data-driven solution for efficient field management.

The introduction of LSTM and GRU architectures has greatly improved the prediction of complex dynamics that were previously challenging for classical techniques—delivering reasonable forecast intervals even in early production phases lacking steady-state history. Their gated-memory design overcomes the difficulty that vanilla RNNs face in learning long-range dependencies, making them particularly appropriate for multi-year well behaviour characterised by cyclic fluctuations and seasonal shut-ins. Nevertheless, these networks require abundant, diverse training data to generalise; when only limited single-well data are available (e.g., new completions), supplementary information from neighbouring wells or static features is often necessary. Their intricate architectures are also sensitive to hyperparameters, which in practice must be tuned via expert knowledge or optimisation heuristics such as PSO and GAs. For example, Qiu et al. [110] proposed a GRU–Kolmogorov–Arnold network (KAN) hybrid model that employs the MissForest algorithm to impute outliers, uses the Pearson correlation coefficient to select salient features, and leverages PSO for automatic hyperparameter tuning, thereby enhancing predictive accuracy and computational efficiency for well-production time series [110]. Zhang et al. [111] developed a GA-optimised GRU–FNN hybrid network for monthly single-well production prediction in high-water-cut reservoirs; the optimisation markedly improved forecast precision.

Given continued increases in computational power and data volume, RNNs and their derivatives (LSTM and GRU) have become mainstream tools for time-series forecasting in oil- and gas-production applications.

3.2 | Sequence-to-Sequence (Seq2Seq) Models

End-to-end Seq2Seq models were first introduced in machine translation. As illustrated in Figure 6, the canonical architecture comprises two RNN sub-networks that act as an encoder and a decoder. The encoder 'compresses' the input sequence into a fixed-length context vector, which the decoder then unfolds to generate the output sequence. Because the framework naturally accommodates variable-length inputs and outputs, as well as long-horizon forecasts, it has been adopted for joint production prediction across multi-well networks. Cui et al. [78] employed a

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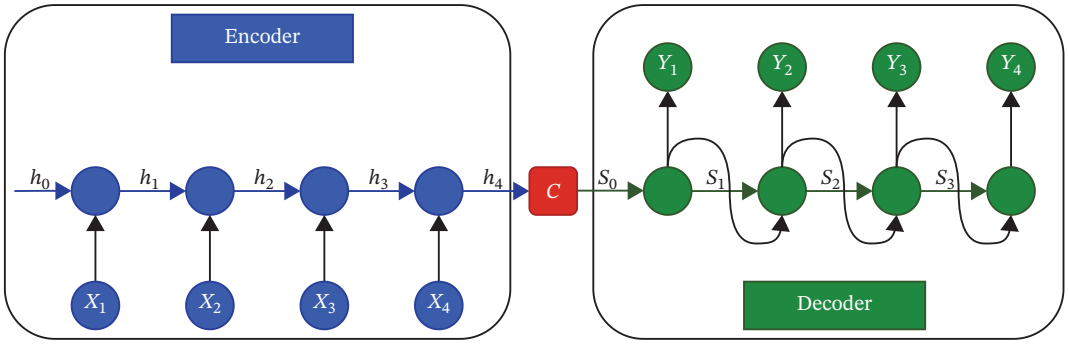

FIGURE 6 | Schematic diagram of the Seq2Seq network architecture.

Seq2Seq–LSTM, encoding neighbouring-well production histories together with inter-well distances and other static attributes to forecast future production at multiple time steps for a target well. To enhance accuracy, the authors incorporated normalised mutual information (NMI) to select features, discarding wells that exhibited weak correlation with the target, and thereby improving the encoder input quality.

A principal advantage of end-to-end models is their flexibility in handling heterogeneous input and output lengths while seamlessly integrating attention mechanisms. Attention allows the decoder to assign different weights to different portions of the encoder output at each prediction step, thereby circumventing the information-bottleneck problem inherent in compressing an entire sequence into a single context vector [112]. In production forecasting, this capability enables the network to learn autonomously which historical moments or which neighbouring wells most influence the current prediction, eliminating the need for manual feature engineering—an especially valuable property for complex production series characterised by multiple perturbations and intermittent shut-in periods. Cho et al.'s [113] seminal work in machine translation first identified that relying solely on the encoder's final context vector leads to information loss for long sequences and proposed an attention mechanism that allows the decoder to 'peek' at different parts of the source sequence during generation. A similar concept has been applied in reservoir forecasting; for instance, Fan et al. [31] integrated an external 'operational event' sequence into an ARIMA–LSTM hybrid, highlighting well shut-in and restart events. This approach enabled the model to capture linear production trends while accommodating non-linear fluctuations, thus, delivering significantly higher predictive accuracy than models that ignore operational interventions [31].

3.3 | Convolutional–Recurrent Hybrid Models

To further enhance forecasting performance, researchers have fused the feature-extraction capacity of convolutional neural networks (CNNs, Figure 7) with the sequence-modelling capability of recurrent networks, thereby creating hybrid DL architectures. CNNs excel at uncovering local patterns and spatial correlations; when applied to time-series data, they can extract motif-like features from contiguous subsequences. Feeding CNN–derived features into an LSTM, GRU or other RNN enables the model to capture both temporal dependencies and non-linear interactions.

For example, Kim and Durlofsky [114] developed a CNN–RNN surrogate to predict oil and water production, demonstrating that the combined architecture markedly outperformed a stand-alone RNN in accuracy. Chen et al. [71] designed a CNN–GRU model for well production during tertiary recovery (EOR) operations; the hybrid achieved superior predictive accuracy and stability on the test set relative to traditional neural network and statistical baselines.

Attention mechanisms have also been incorporated to reinforce hybrid models. Pan et al. [59] proposed a CNN–LSTM architecture augmented with self-attention (CNN–LSTM–SA) for field-scale production forecasting. In this framework, the CNN extracts spatial features, the LSTM captures temporal dependencies and the self-attention module dynamically emphasises critical moments and factors along the time axis, yielding substantially higher accuracy than classical machine learning methods [59]. Zhen et al. [46] introduced a forecasting network that combines a temporal convolutional network (TCN; Figure 8) with an attention module. By identifying and focusing on the most influential time steps, the TCN–attention model delivered significantly improved accuracy over pure TCN, LSTM and vanilla RNN benchmarks across multiple well datasets [46, 115].

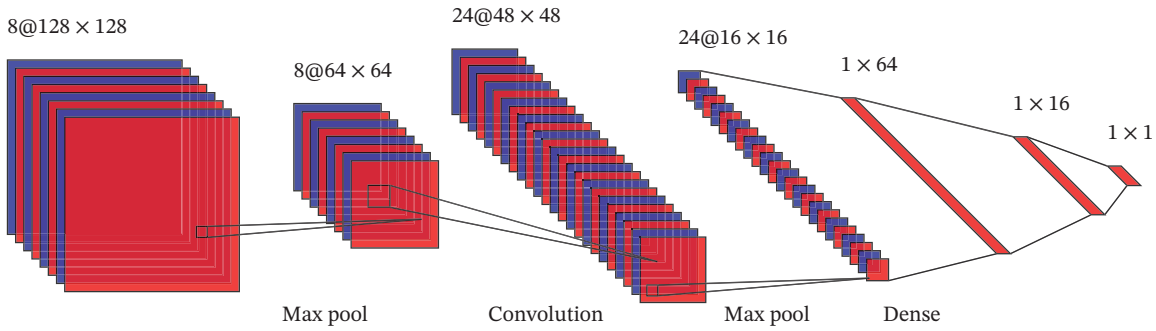

FIGURE 7 | Schematic diagram of the CNN architecture.

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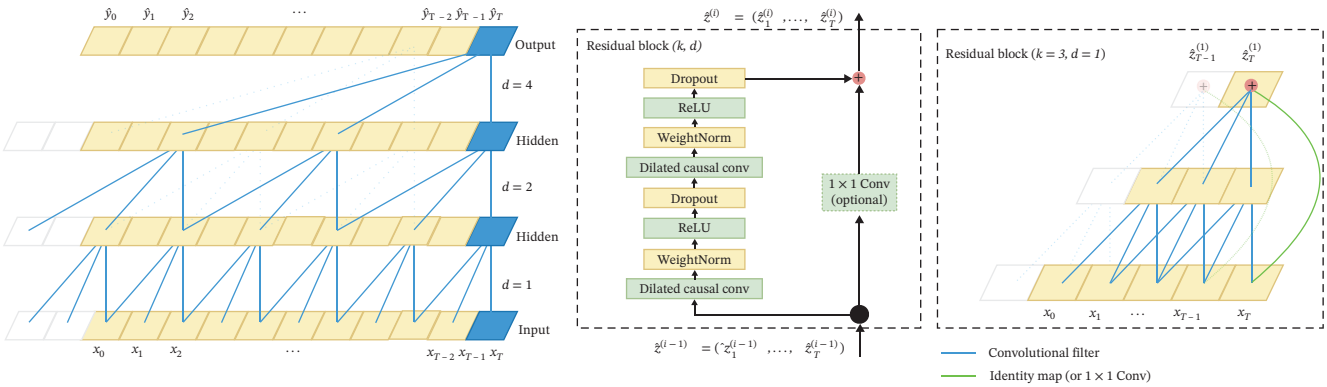

FIGURE 8 | Schematic diagram of the TCN architecture [115].

Overall, CNN–RNN hybrids leverage the complementary strengths of different neural network families and have become an effective means of boosting predictive precision in oil- and gas-production applications. Across diverse reservoir datasets, these composite architectures consistently achieve lower errors and higher reliability than single-model counterparts.

3.4 | Transformers and Emerging Architectures

In recent years, attention-based sequence models—most notably transformers—have achieved remarkable success across numerous disciplines [116] and are now being explored for oil- and gas-production forecasting. As shown in Figure 9, a transformer dispenses with the recurrent structure of RNNs; instead, it relies on a self-attention mechanism that enables fully parallel computation over an entire sequence. The canonical model adopts an encoder–decoder stack in which each layer comprises multi-head self-attention, position-wise feed-forward networks, residual connections and layer normalisation. Positional information is injected via sinusoidal encodings so that ordering relationships are retained even in the absence of recurrence.

Abdrakhmanov et al. [117] developed a transformer-based deep neural network for long-horizon prediction of transient well-production data. The transformer significantly outperformed

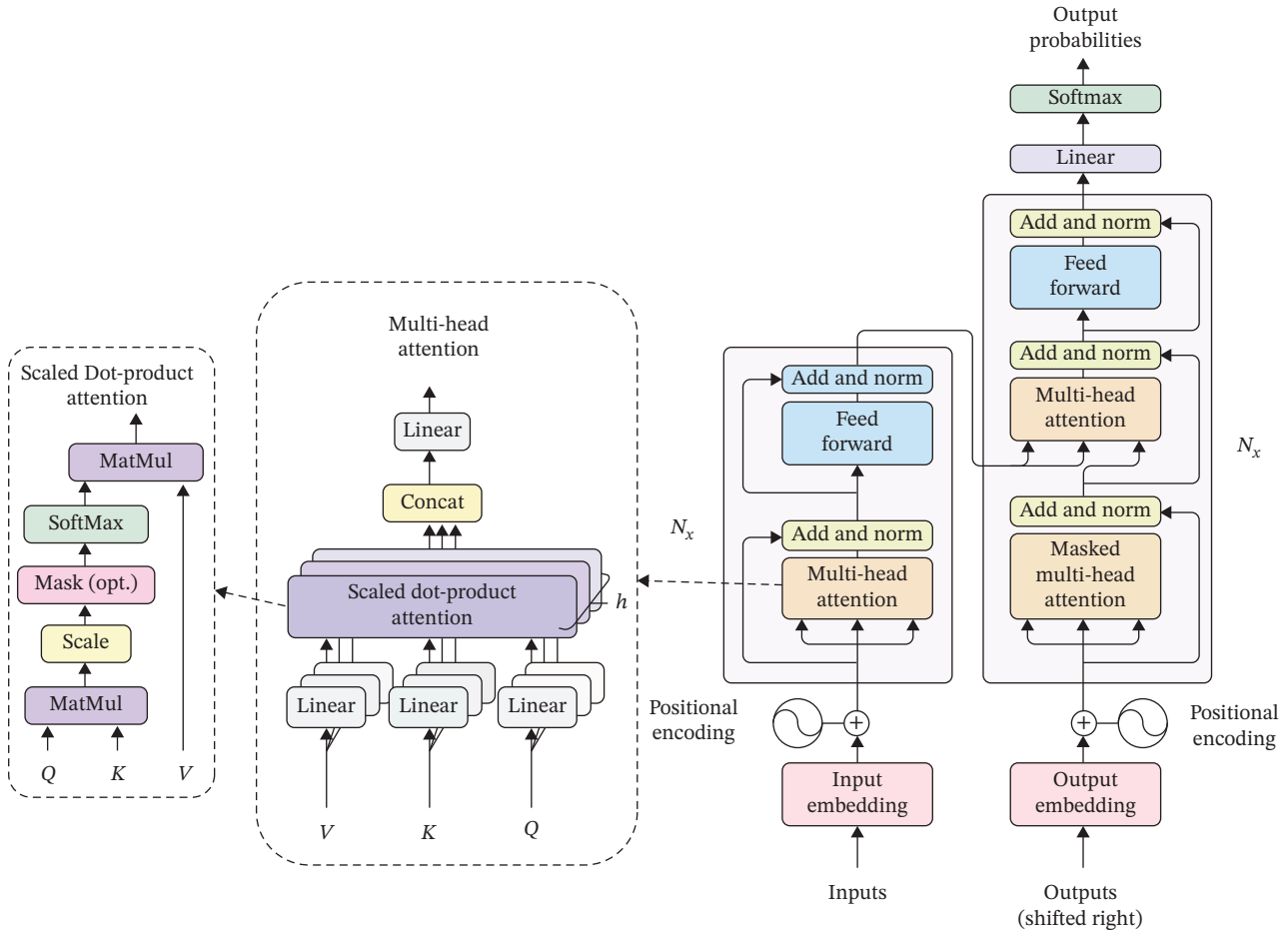

FIGURE 9 | Schematic diagram of the transformer architecture [116].

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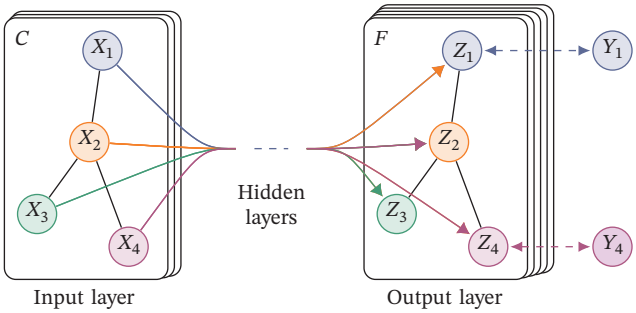

FIGURE 10 | Schematic depiction of multi-layer graph convolutional network (GCN) for semi-supervised learning with C input channels and F feature maps in the output layer. The graph structure (edges shown as black lines) is shared over layers, labels are denoted by Y_i [120].

conventional LSTM and GRU baselines. Leveraging transfer learning, the authors pre-trained the network on one well and then fine-tuned it for a target well, thereby markedly improving accuracy on the latter. They also constructed a global transformer that ingests historical data from multiple wells simultaneously to capture inter-well interference, successfully simulating the evolution of well-cluster productivity [117]. This study demonstrates the transformer's potential for modelling field-scale spatio-temporal patterns and complex multi-well coupling behaviour.

Carpenter [118] proposed a transformer-based deep surrogate for forecasting well performance. Using a sliding-window technique, multi-variate time-series inputs—bottom-hole pressure, production rate, temperature, surface parameters and others—were converted into training samples for the model. In comparative tests, the transformer more accurately captured non-linear dependencies and transient variations than LSTM– or GRU–based networks [118].

To address the challenges posed by non-stationary field data, Kumar [119] introduced a non-stationary transformer framework featuring 'series-stationarisation' and 'de-stationary attention' modules. On long-term, multi-variate, field-level production-forecast tasks, this architecture delivered substantially higher modelling accuracy than the vanilla transformer [119].

Beyond the transformer framework, researchers have also begun to investigate other innovative architectures. For instance, time-series models that incorporate graph neural networks (graph convolutional networks [GCNs]; Figure 10) [120] can explicitly encode well-network topology and injector-producer relationships. Du et al. [51] coupled a GCN with a GRU to create a DL model for inter-well connectivity calibration and production-performance forecasting. By exploiting dynamic production data, the model autonomously refined the initial connectivity—originally derived from permeability, porosity and well-spacing metrics—and accounted for the time-varying influence of reservoir conditions, reducing predictive error by 78.6 % [51].

Another example is the KAN (Figure 11) [121], derived from the Kolmogorov–Arnold representation theorem and capable of capturing highly non-linear relationships in well-production data. In Hu et al.'s [122] TCN–KAN forecasting framework, the KAN component proved pivotal, markedly enhancing the model's ability to track the non-linear production trajectories of individual wells and driving the overall improvement in predictive accuracy.

These exploratory studies signal an emerging trend toward leveraging advanced DL architectures—such as GCNs and KANs—for oil- and gas-production forecasting. Preliminary results confirm their feasibility and effectiveness in attaining higher accuracy under increasingly complex reservoir scenarios.

In summary, a broad spectrum of machine learning techniques has been applied to well-productivity forecasting, each delivering tangible successes. Shallow models—including SVR and RFs—remain efficient and reliable when data volumes are modest. DL

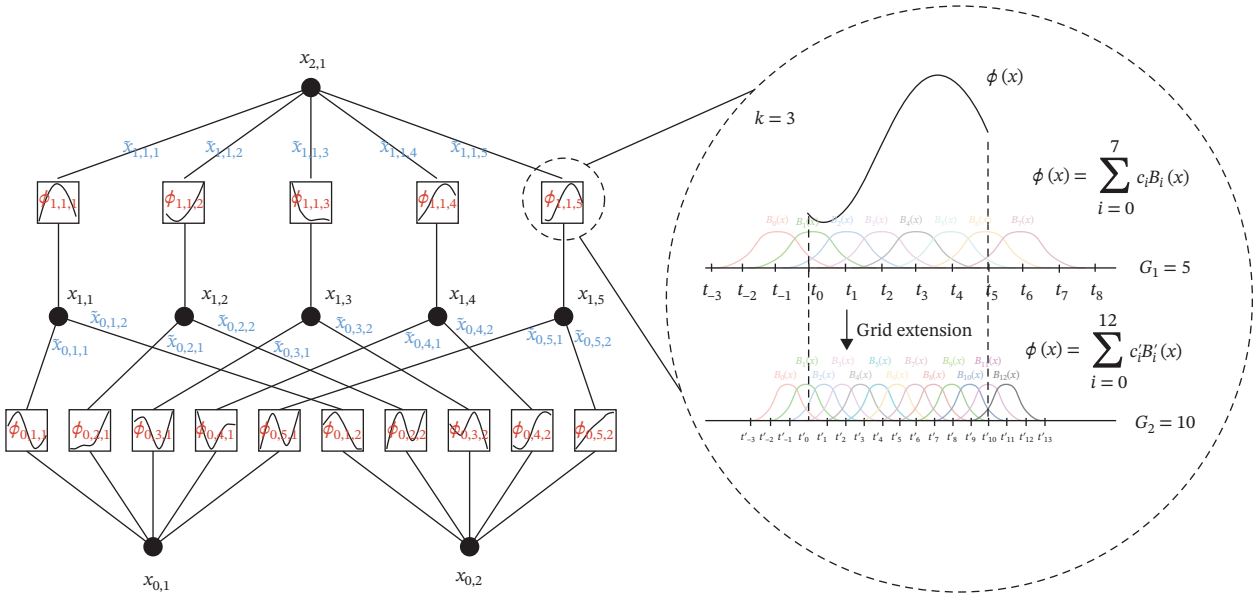

FIGURE 11 | Schematic diagram of the Kolmogorov–Arnold network (KAN) architecture, the KAN is a novel neural network paradigm that relocates learnable activation functions from nodes to edges. Traditional linear weights are replaced with parameterised B-splines (right panel) and the model dynamically switches between coarse- and fine-grained grids to achieve both computational efficiency and high-accuracy approximation. Its edge-based activation propagation mechanism (left panel), together with the network's adaptive design, markedly enhances performance in scientific-computing applications [121].

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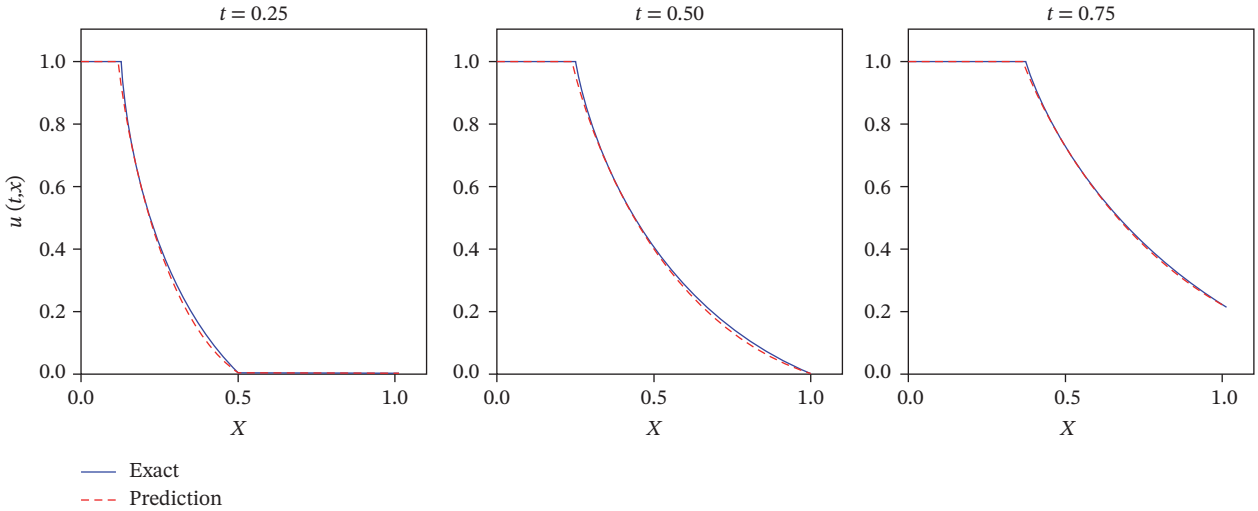

FIGURE 12 | Comparison of the predicted by the neural network and the exact solutions of the PDE with concave flux function, corresponding to the three different times $t = 0.25$, 0.5 and 0.75, where x denotes the dimensionless spatial coordinate and u represents the water saturation S_w [126].

architectures such as LSTM and GRU excel at capturing long-range dependencies, offering superior accuracy for production time-series prediction. Hybrid frameworks that blend CNNs, recurrent networks and attention mechanisms achieve further performance gains, often setting state-of-the-art benchmarks in specific case studies. Emerging models, most notably the transformer family, show distinct promise for handling complex and multi-well systems.

Importantly, the optimal choice of method depends on data characteristics and problem scale. A simple algorithm may suffice for short-term forecasts at the single-well level, whereas long-horizon and coupled predictions across numerous wells generally demand the expressive power of advanced DL architectures.

Despite their successes, both traditional machine learning and DL approaches share a critical shortcoming: limited interpretability. These models are typically treated as 'black boxes' whose internal mechanics and decision pathways remain opaque. Such lack of transparency poses a significant hurdle in disciplines—like petroleum engineering—where high levels of trust and verifiability are mandatory.

To address this challenge, PINNs have emerged, integrating data-driven learning with explicit physical constraints. The sections that follow therefore focus on the use of PINNs in production forecasting, examining current applications and outlining prospects for future development in this domain.

4 | Physics-Informed Machine Learning (PIML)

PIML refers to a class of algorithms that embed physical laws as explicit constraints within data-driven models. Although early precursors can be traced to the 1990s, the concept gained momentum after Raissi et al. [123] proposed the PINN framework in 2017. The key idea is to incorporate the governing partial differential equations (PDEs) of multi-phase flow, along with initial and boundary conditions and other physical priors, directly into the neural network training process by adding a physics-residual term to the loss function [123]. During optimisation, the network is therefore penalised whenever its predictions violate the underlying physics, producing solutions that honour both the data and the PDEs [124]. This approach markedly enhances physical fidelity and extrapolation capability, enabling the model to capture essential trends even when observations are sparse or noisy.

PIML research has flourished in petroleum applications. Foundational contributions include Raissi et al.'s [123] original PINN formulation, which introduced DL as a general solver for PDEs [124]. Cai et al. [125] subsequently offered a systematic review of physics-driven DL frameworks, demonstrating that PINNs can achieve accuracy comparable to traditional numerical methods in fluid-mechanics problems while exhibiting superior robustness when confronted with incomplete data—an insight that laid the groundwork for extending PINNs to more intricate engineering systems such as production forecasting.

In reservoir engineering, the group led by Prof. Tchelepi at Stanford University ranks among the earliest adopters. They applied PINNs to the Buckley–Leverett equation for immiscible two-phase displacement, incorporating an appropriate numerical-diffusion term to guide the network toward physically stable saturation profiles. The study proved that PINNs can directly solve highly non-linear hyperbolic equations that conventionally require artificial viscosity for stabilisation. Figure 12 shows that the computed saturation distributions remain physically consistent [126].

Building on these seminal works, a growing body of literature now leverages PIML as a core methodology for well-productivity forecasting and reservoir simulation, signalling a decisive shift toward physics-constrained data analytics in petroleum science and engineering.

4.1 | Transformers and Emerging Architectures

Li et al. [127] proposed a theory-guided graph neural network (TGNN) that embeds high-fidelity numerical-simulation data, fundamental physical laws, boundary conditions and expert heuristics directly into the loss function, thereby solving two-phase-flow problems at reservoir scale. Addressing the strong non-linearity introduced by source-sink terms in single-phase flow,

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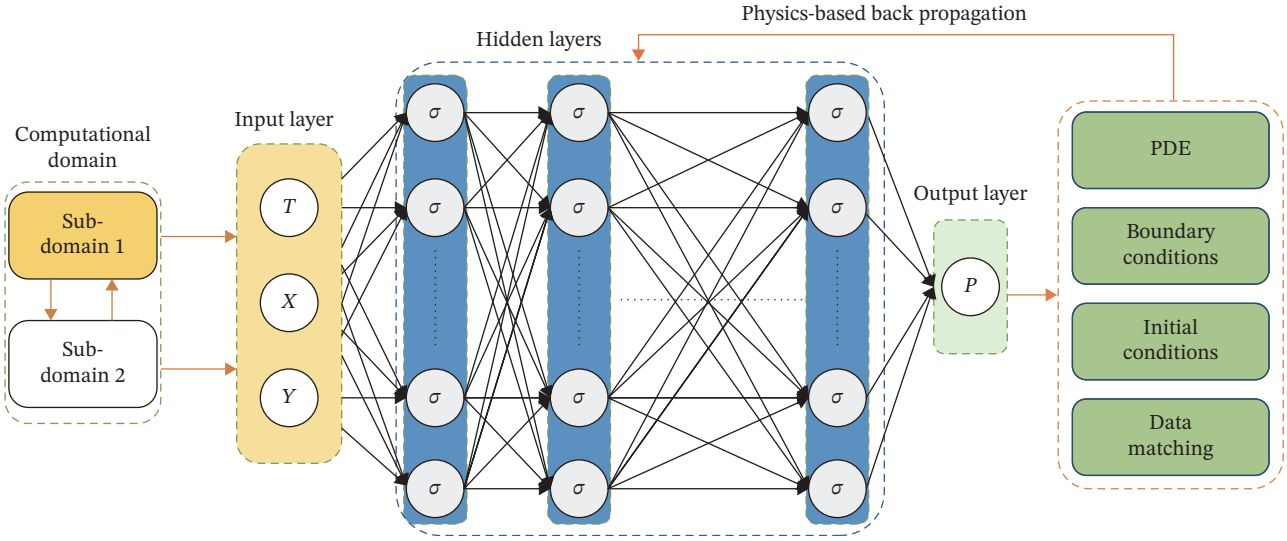

FIGURE 13 | The structure of the PINN–DD model [130].

Daolun et al. [128] designed an improved physics-constrained DL algorithm that explicitly incorporates latent PDE features into the loss to mitigate steep gradient variations, successfully solving flow equations with well sources. Yan et al. [129] developed an enhanced PINN that couples the explicit-fracture discrete-fracture model (EDFM) with a finite-volume scheme, enabling high-accuracy, label-free simulation of two-phase flow in both matrix and fracture domains for two- and three-dimensional settings.

Han et al. [130] introduced a domain-decomposition strategy for large-scale reservoir simulations containing wells, termed PINN–DD (architecture shown in Figure 13). The computational domain is partitioned into two sub-regions—one encompassing the wellbore and one without wells—each solved by a separate PINN and coupled at the well interface. Internal boundary conditions near the well are imposed via the Peaceman formulation, allowing full-field pressure-field reconstruction from sparse data consisting only of wellhead production rates and limited bottom-hole pressure measurements. As illustrated in Figure 14, handling the singular pressure gradient near the well markedly improves simulation accuracy, with the resulting pressure field achieving an R^2 of 0.98 [130].

Bi et al. [131] proposed a physics-informed spatio-temporal neural network (PI–STNN) that embeds fluid-flow mechanisms into a DL framework for reservoir simulation. Architecturally, the model couples a convolutional encoder-decoder network with a CNN–LSTM module, allowing it to learn, in a data-driven manner, the spatial distribution and temporal evolution of reservoir pressure fields; production forecasts are subsequently obtained by integrating the Peaceman well-flow formulation. Crucially, the training objective augments the data-misfit term with a physics-residual term enforcing the conservation laws that govern multi-phase flow, so that the network simultaneously minimises observational error and physical inconsistency.

In comparative tests against an otherwise identical but purely data-driven network (STNN), PI-STNN delivered markedly superior accuracy—particularly in highly heterogeneous and naturally

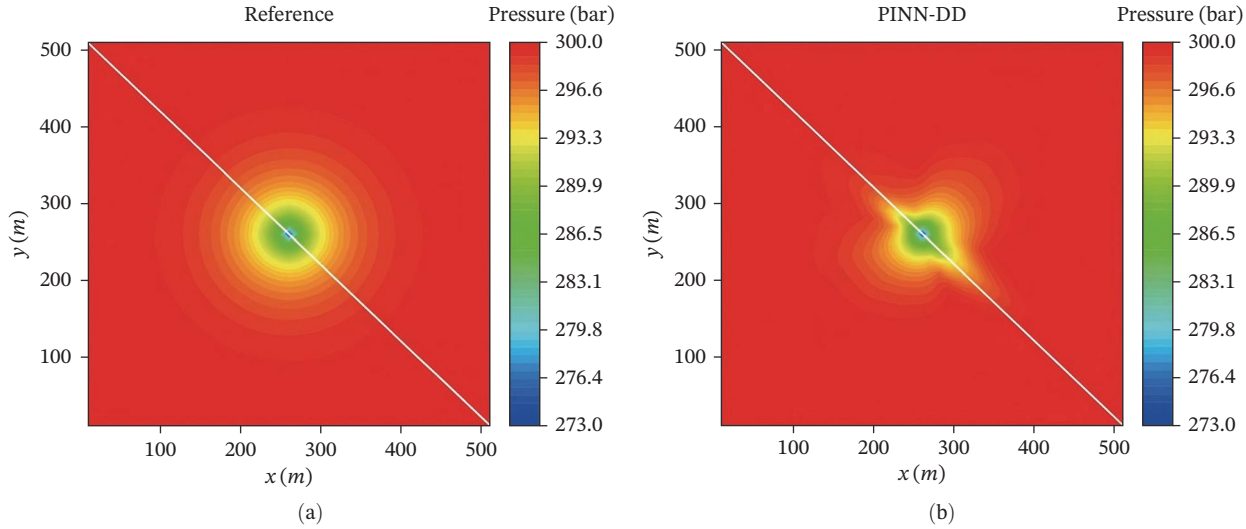

FIGURE 14 | (Continued)

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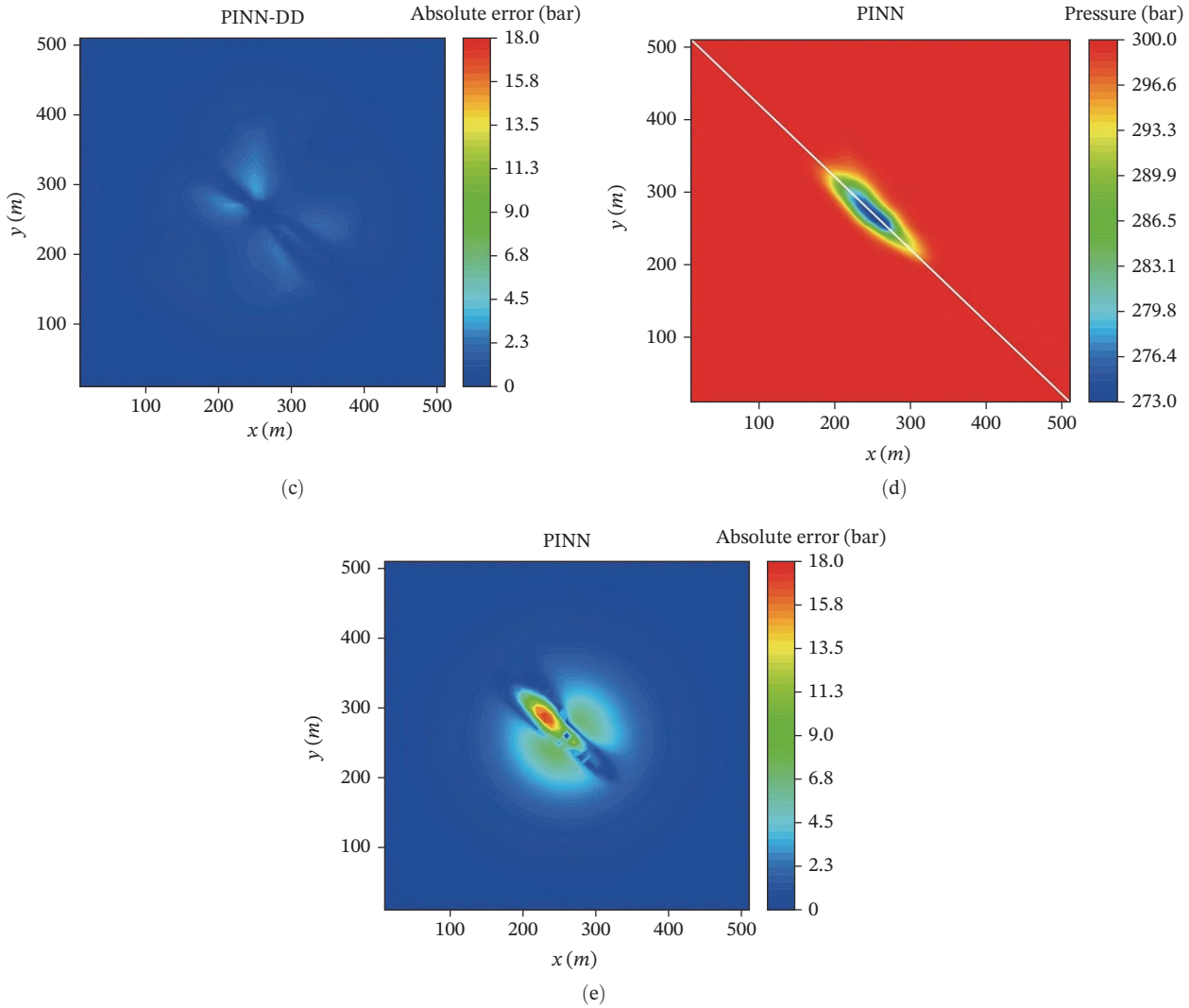

FIGURE 14 | Pressure fields of numerical simulation reference (a), PINN–DD (b, c) and PINN (d, e) at $t = 30$ days [130].

fractured reservoirs—maintaining high-fidelity predictions of pressure and saturation and exhibiting robust generalisation on the test set (Figure 15). By contrast, the STNN failed to generalise in such complex settings, underscoring the advantages conferred by the incorporation of physical priors and the CNN–LSTM's capacity to capture temporal dynamics that conventional models struggle to learn autonomously.

4.2 | Production-Data Forecasting

Cornelio et al. [132] transferred the PINN philosophy to the prediction of unconventional-reservoir production by first obtaining a baseline solution from conventional numerical simulation and then training a neural network to learn the residual between the simulator output and the field data. This residual-learning strategy marries the strengths of physics-based and data-driven approaches, delivering higher accuracy than stand-alone simulators while retaining sensitivity to parameter variations in shale-oil wells [132]. The workflow exemplifies a practical 'numerical-simulation + machine learning' hybrid that can be adopted in field operations.

Li et al. [133] proposed a bidirectional GRU (BiGRU)–deep hybrid neural network (DHNN) hybrid model that integrates static geological attributes (fracture parameters and lithology), dynamic production records and spatio-temporal information as physics-motivated constraints. By combining a BiGRU with a DHNN, the framework achieved long-term production forecasts for multiple fractured wells with superior accuracy, robustness and generalisation relative to conventional RNN, LSTM and GRU benchmarks [133].

Ma et al. [134] introduced a GRU–MLP hybrid in which physical-constraint parameters—such as inter-well spacing, number of frac stages and completion variables—are supplied as feature inputs. The added physics consistency improved predictive precision and stability for shale-gas wells, particularly in long-horizon forecasts, outperforming purely data-driven models [134].

Wang et al. [135] developed a physics-enhanced CNN–LSTM–AM architecture (Figure 16). Convolutional layers capture spatial patterns, the LSTM layer learns temporal dynamics and an attention module focuses on critical state changes. By filtering input features and embedding the five-line flow model for shale oil directly into the loss function, augmented with Bayesian optimisation and SHAP interpretability analysis, the model delivered high-accuracy forecasts that respect flow physics. As illustrated in Figure 17, the CNN–LSTM–AM predictions align with reservoir-flow theory and suppress non-physical artefacts [135].

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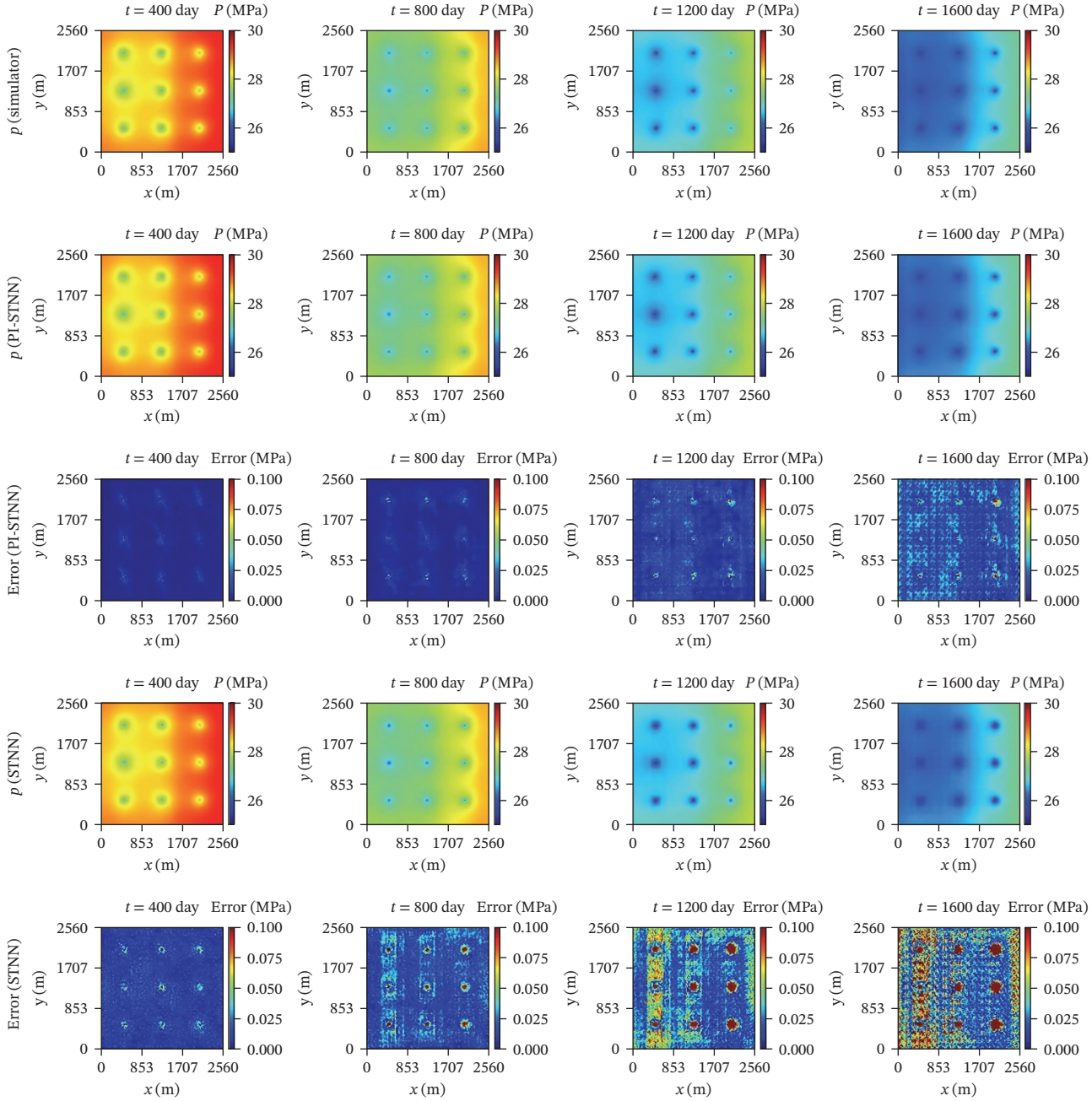

FIGURE 15 | Pressure fields and errors at four time points. It provides a comparative view of both the full-field pressure calculations performed by the simulator, PI-STNN and STNN at four distinct time points, and the absolute differences between the surrogate model predictions and their corresponding simulator values [131].

To improve both accuracy and physical plausibility in shale-gas forecasts, Wang et al. [135] incorporated Caputo fractional-order derivatives into a PINN, enabling the network to learn the intricate non-linear decline behaviour characteristic of unconventional wells. The fractional PINN embeds sparse physical priors within training and extends the framework's capacity to model strongly non-linear decline curves, markedly enhancing physical consistency [136].

Numerous studies have shown that integrating empirical decline-curve formulas with neural networks can markedly enhance predictive performance in data-sparse settings without materially increasing model complexity. Chen et al. [136], for example, used a classical decline-curve model as a front-end module whose output feeds a data-driven neural network; the hybrid framework achieved a mean error of only 14.1 % in long-term production forecasts for tight-gas wells in the Ordos Basin, a substantial reduction relative to a purely data-driven model. Xue et al. [137] embedded the decline-curve analysis (DCA) formulation directly into the loss function of an LSTM network as a physics-based penalty term; by jointly minimising data-fit and physical-residual errors during training, the model lowered long-term production-forecast errors by approximately 42.9 % for tight-gas reservoirs and 13.7 % for carbonate-gas reservoirs, as illustrated in Figure 18. Busby combined a neural network with the Arps decline-curve model, introduced an automatic weighting scheme to balance physical and data losses and incorporated an uncertainty-

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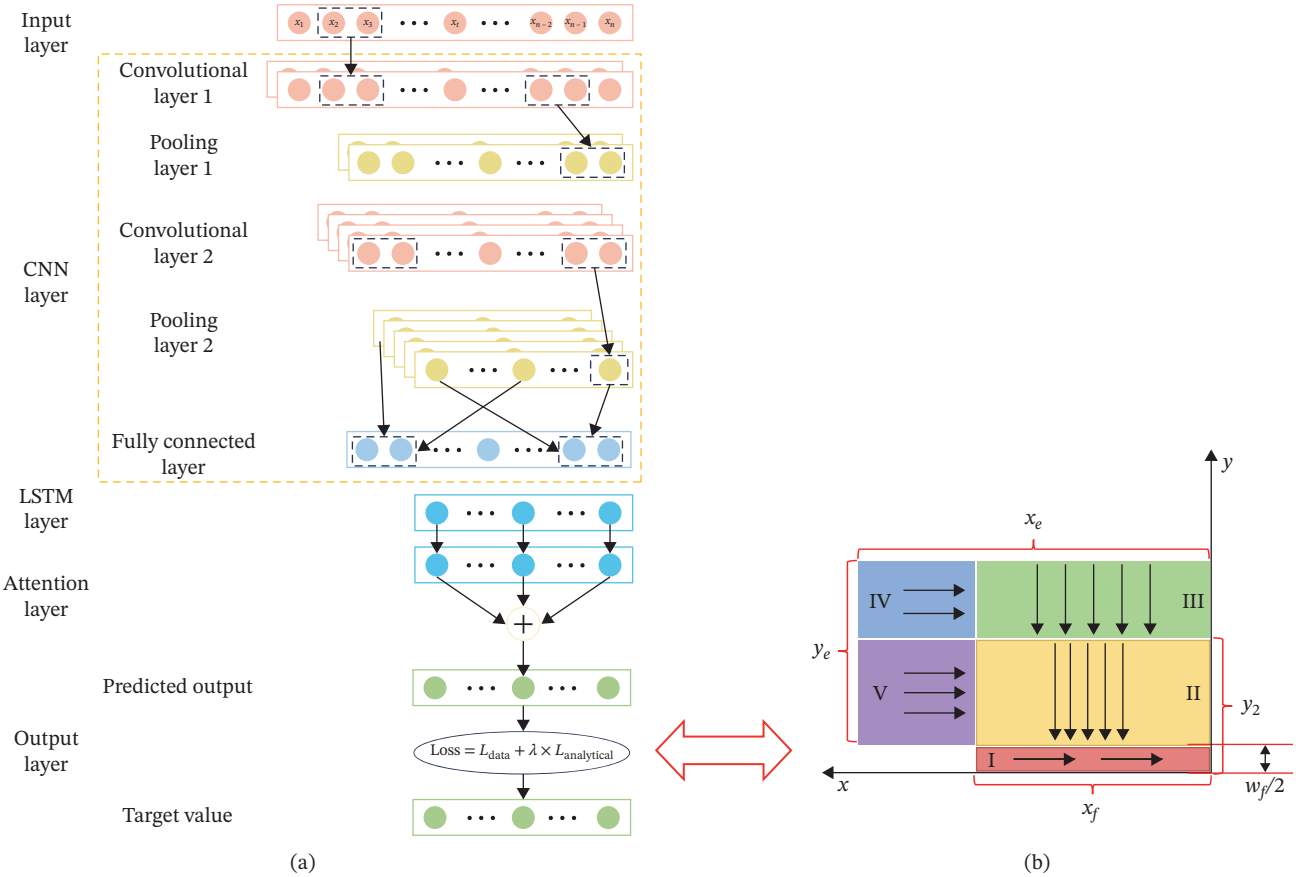

FIGURE 16 | (a) Schematic of a CNN–LSTM–AM model enhanced with physical constraints. (b) Five-linear flow diagram, the hydraulic fracture zone (I), the fracture zone (II) and the non-fractured zone, which is composed of three linear flow regions (III, IV and V). In this model, the y-axis is aligned with the horizontal well, the x-axis extends along the fracture, x_f is the half-length of a fracture and w_f is the width of the fracture. The fractures have a spacing of 2 y_e and y_2 is the distance between the end of Zone II and the fracture. Additionally, x_e is the distance from the tops end of Zones IV and V to the wellbore. Finally, the model considers the fracture height and the reservoir thickness h [135].

quantification module. The resulting workflow enables automated, multi-segment, probabilistic long-term forecasting for oil wells and is demonstrably more robust than conventional decline-curve methods [138].

4.3 | Advances and Variants of PINN Techniques

As research has progressed, numerous enhancements to the PINN framework have been proposed to improve both training

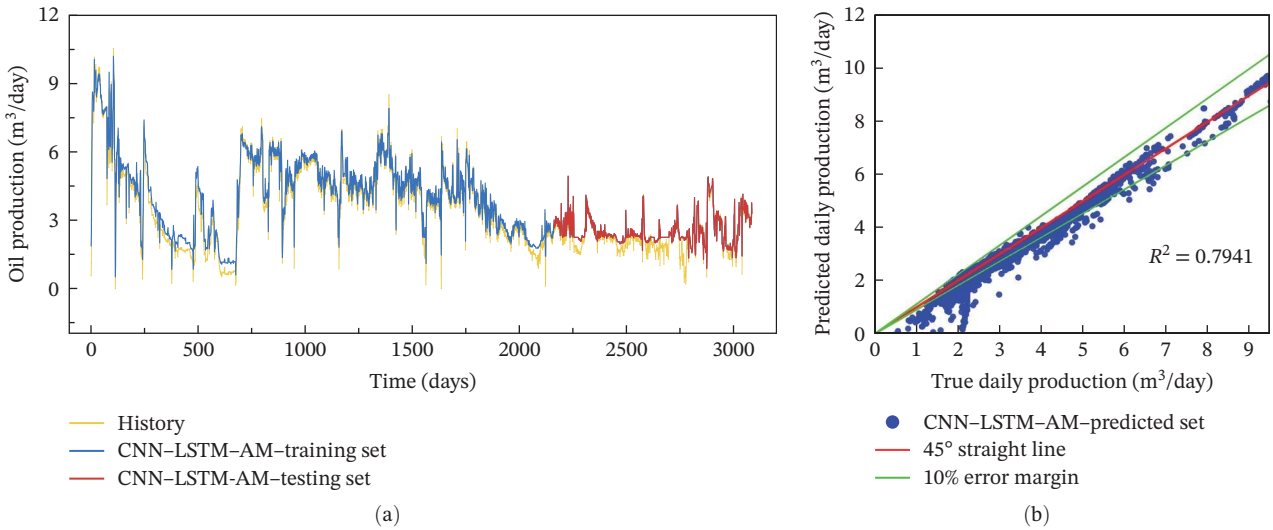

FIGURE 17 | (Continued)

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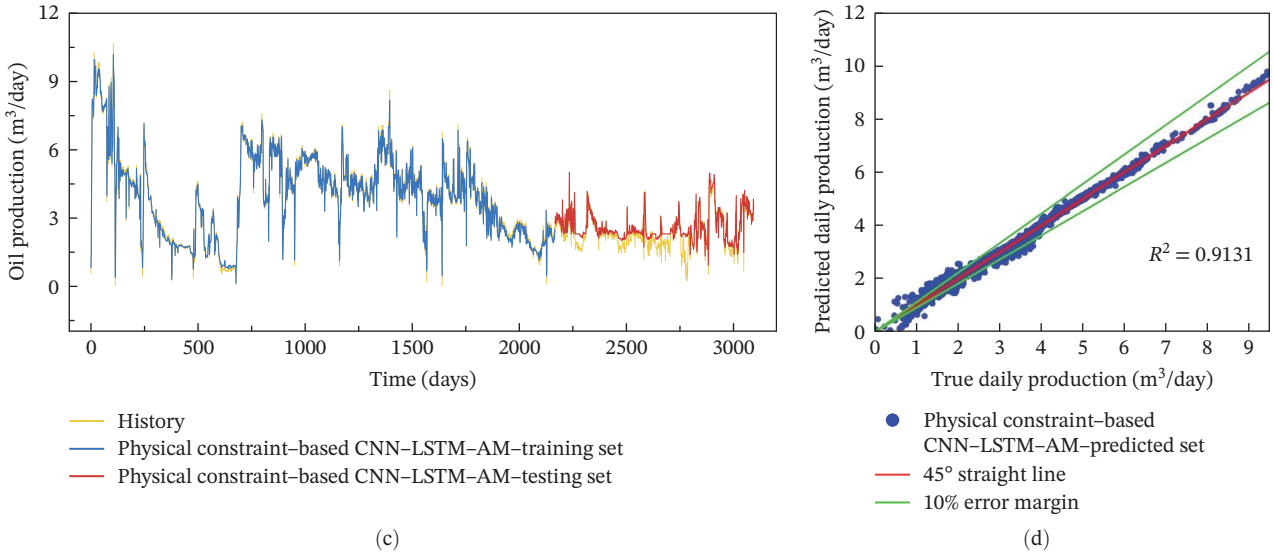

FIGURE 17 | Prediction effects of the three machine learning production prediction models for oil well. Prediction results of the (a) CNN–LSTM–AM. (c) Physical constraint–based CNN–LSTM–AM. Error analysis of the shale oil production prediction models (b) CNN–LSTM–AM. (d) Physical constraint–based CNN–LSTM–AM [135].

efficiency and predictive accuracy. Shukla et al. [139] introduced the XPINN approach, a parallel domain-decomposition strategy that partitions the computational domain into multiple spatial and temporal sub-domains and trains independent yet interconnected neural networks for each sub-region, thereby achieving substantial speed-ups through concurrency. Chiu et al. [140] developed the CAN–PINN framework, which combines the strengths of automatic differentiation and numerical differencing, markedly accelerating PINN–based PDE solvers.

In petroleum applications, Song et al. [141] proposed the DPIE–HPK workflow. The method first employs an LSTM to forecast well production over future time windows, then merges these forecasts with historical data via a physics-informed data-supplementation (PIDS) procedure. Coupled with kriging and other physics-based techniques, DPIE–HPK yields high-accuracy predictions of both field-wide production and residual-oil saturation while maintaining excellent deployment efficiency and applicability [141].

Zhang [142] introduced a physics-informed convolutional neural network (PICNN) that integrates the Darcy flow equations—discretised by the finite-volume method—directly into the loss function. The network performs high-fidelity and unsupervised simulations of transient flow in heterogeneous reservoirs, relying solely on PDEs, boundary conditions and initial conditions without labelled data, thus, offering a practical solution for data-sparse reservoirs [142].

Jin et al. [143] modified an end-to-end LSTM architecture by embedding a physics encoder and an attention mechanism, yielding the PIED model. PIED extracts physics-relevant features from high-dimensional fracturing parameters and focuses on the information most critical to production, achieving higher accuracy on

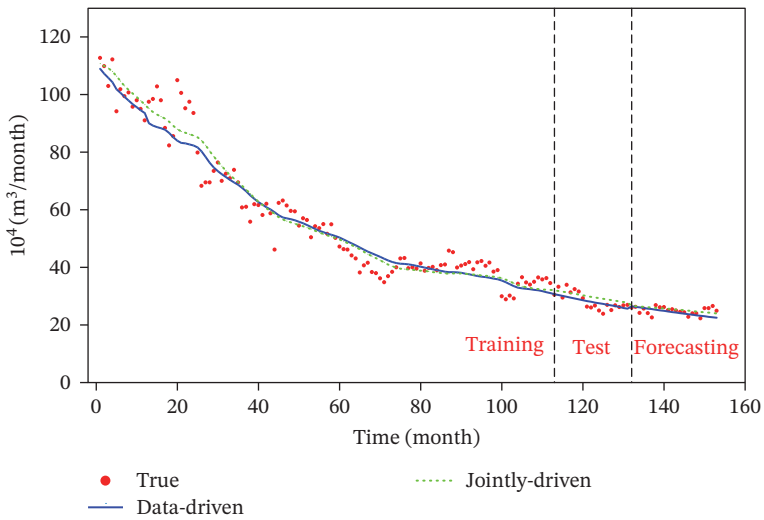

FIGURE 18 | Comparisons of prediction results based on data-driven and jointly driven LSTM models for well, the MAPE of the data-driven LSTM model is 8.14%, while the MAPE of the jointly-driven LSTM model is 4.85% [137].

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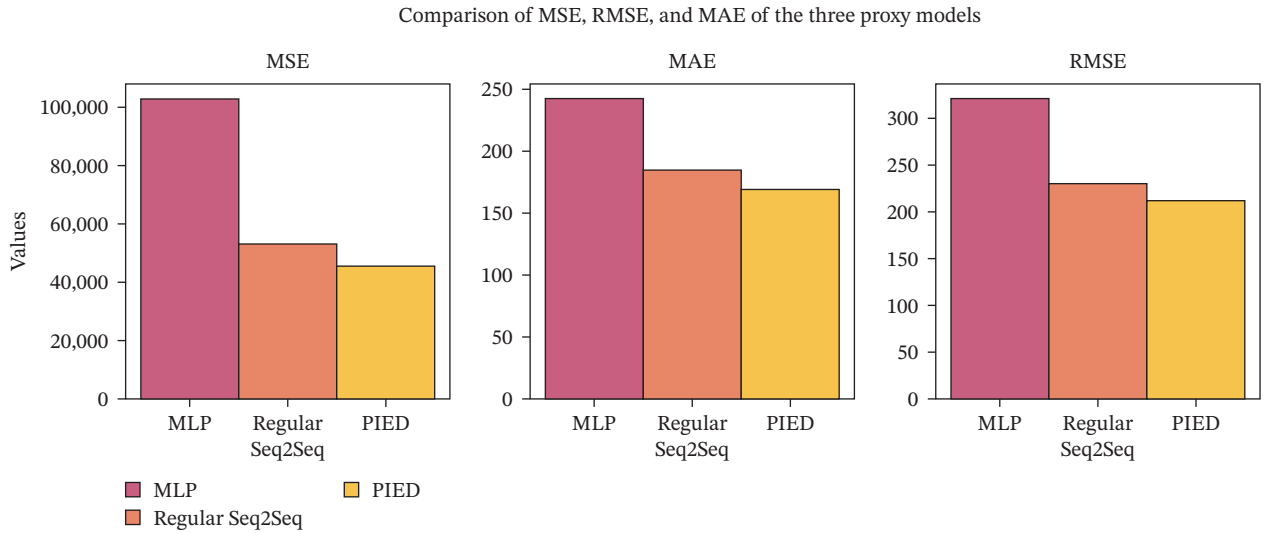

FIGURE 19 | The comparison of MSE, RMSE and MAE of the three models [143].

limited datasets. A performance comparison with conventional data-driven models is provided in Figure 19 [143].

4.4 | Efficient and Accurate Proxy-Model Construction

Proxy models represent the most fundamental and widespread application of PIML in petroleum engineering. Their primary objective is to replace computationally expensive full-physics simulators, and thus, enable rapid decision making. Whereas a single run of a conventional numerical simulator may require hours or even days, an already-trained proxy typically delivers predictions within seconds—or less—yielding speedups of several orders of magnitude.

Traditional proxy techniques, such as polynomial response surfaces or kriging interpolation, can handle only low-dimensional input–output relationships, and therefore, struggle with the high-dimensional character of reservoir problems. For example, they cannot readily ingest an entire permeability field (a high-dimensional matrix) and forecast the spatio-temporal evolution of saturation. By contrast, PINN–based proxy models employing DL architectures—particularly CNNs and RNNs—are well suited to such high-dimensional spatio-temporal data. Zhong et al.'s [25] Co-GAN proxy, for instance, accepts both reservoir properties (e.g., permeability fields) and time as inputs and simultaneously predicts pressure and saturation distributions, thereby providing a comprehensive description of reservoir dynamics. Similarly, Coutinho et al. [144] proposed the E2CO proxy framework, which integrates physics constraints, auto-encoder-based dimensionality reduction and joint control/output prediction. E2CO efficiently and accurately emulates reservoir-state parameters along with wellhead production rates and pressures; as illustrated in Figures 20 and 21, the framework achieves prediction accuracies

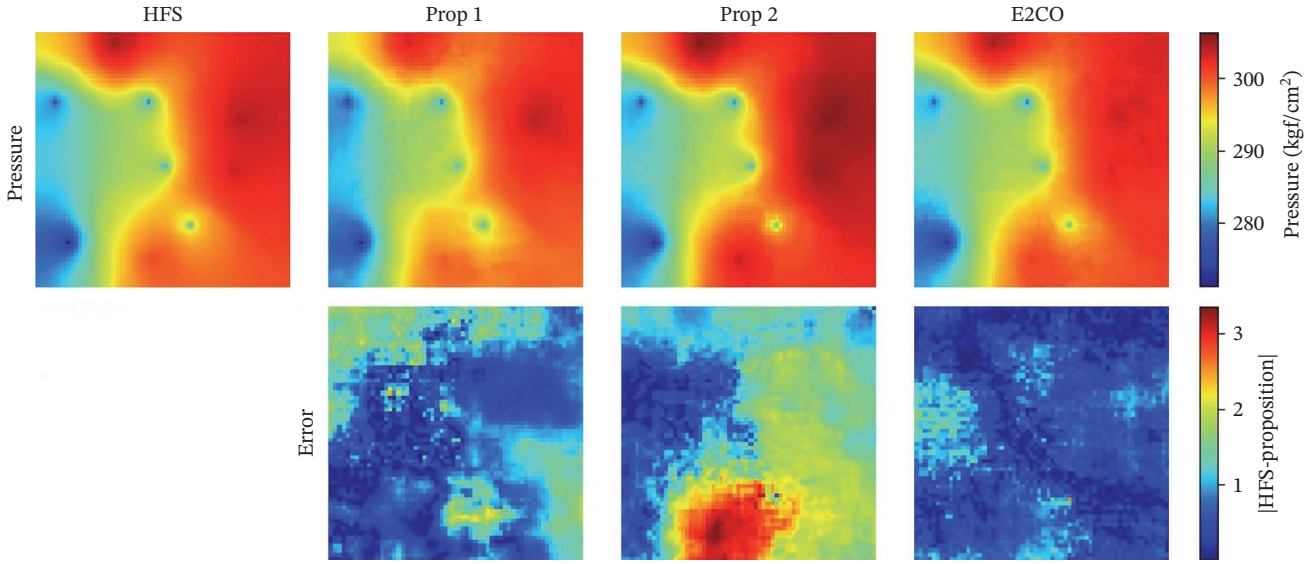

FIGURE 20 | Pressure-field predictions from four models, HFS represents the high-fidelity simulator and serves as the reference solution, Prop1 is the proxy model obtained with the E2C-ROM approach [145], Prop2 is derived from E2C-ROM with an additional well-data loss term, E2CO, embed to control and observe, an alternative approach to calculate the output, by having a specific neural network to capture the relation between reduced state, controls and outputs [144].

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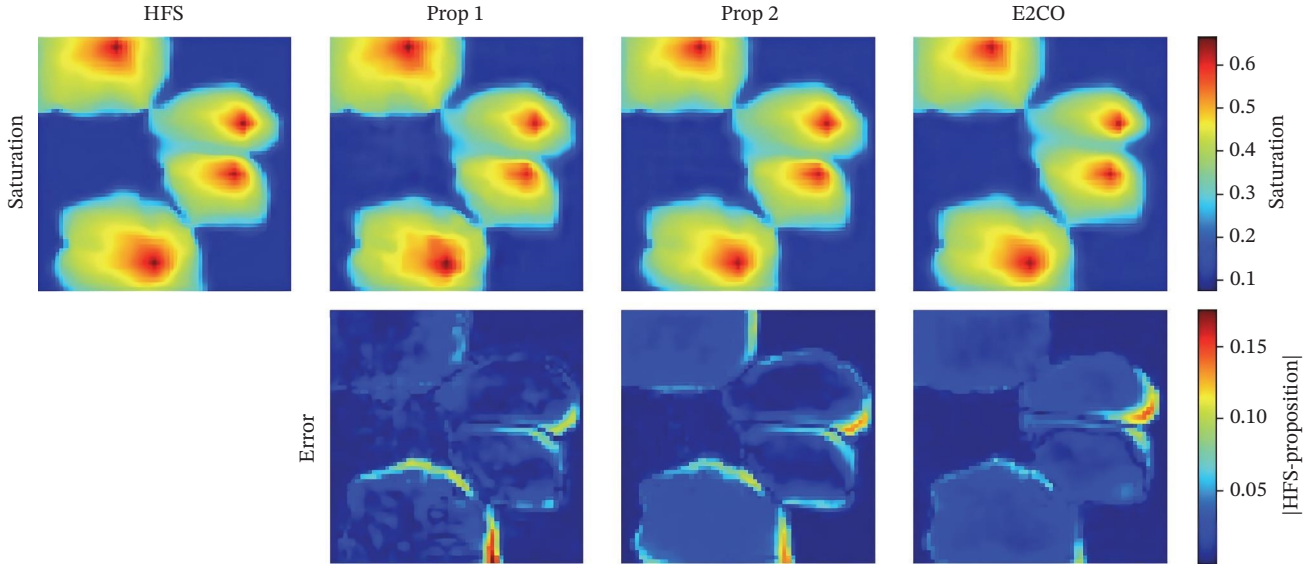

FIGURE 21 | Water saturation field predictions from four models, HFS represents the high-fidelity simulator and serves as the reference solution, Prop1 is the proxy model obtained with the E2C-ROM approach [145], Prop2 is derived from E2C-ROM with an additional well-data loss term, E2CO, embed to control and observe, an alternative approach to calculate the output, by having a specific neural network to capture the relation between reduced state, controls and outputs [144].

comparable to full-physics simulators while operating at dramatically lower computational cost [144].

The foremost advantage that PINN–based proxy models confer is physical consistency. A purely data-driven proxy is essentially a statistical fit to the input–output behaviour of a numerical simulator and has no understanding of the underlying physics. When the input parameters fall outside the training domain, such a model may produce predictions that are physically untenable. In contrast, a PINN proxy embeds governing equations directly into its loss function, ensuring that its outputs comply—at least approximately—with the laws of fluid flow. Consequently, a PINN proxy is not merely 'imitating' the simulator; it is, in effect, learning the physics that the simulator encodes.

This built-in consistency makes PINN proxies markedly more robust, especially in extrapolative and optimisation scenarios. They are, therefore, viewed not just as accelerators for conventional simulators but as a cornerstone technology for constructing digital twins of reservoirs. A high-fidelity PINN proxy can serve as a fast, trustworthy and interpretable virtual reservoir on which engineers can perform a range of 'what-if' analyses and test alternative development strategies without enduring the lengthy runtimes of full-physics simulation, thereby enabling faster and more intelligent reservoir management.

4.5 | Assisted History Matching

History matching is a pivotal inverse problem in reservoir modelling whose objective is to adjust uncertain parameters—such as permeability, saturation and porosity fields—so that simulated outputs (oil, gas and water rates, bottom-hole pressures, etc.) reproduce recorded production data. A satisfactorily matched model is considered reliable for forecasting and development optimisation.

The task is notoriously difficult and ill-posed for two main reasons. First, non-uniqueness: limited observations and extremely high-dimensional parameter spaces allow many dissimilar parameter sets to fit the historical data equally well. Second, computational cost: conventional workflows rely either on engineer-guided trial-and-error or on automated optimisation, both of which require innumerable full-physics simulations and can consume months.

PINNs offer a powerful remedy to both challenges. By encoding the governing flow equations directly into the loss function, PINNs exploit sparse and noisy production data, while preserving physical interpretability and dramatically improving computational efficiency. When a PINN proxy replaces the full-physics simulator, optimisation algorithms can explore vast parameter combinations in minutes rather than months, a workflow termed assisted history matching.

Etienam et al. [146], for example, substituted a highly efficient proxy—the physics-informed neural operator coupled with a cluster-classify-regress expert mixture (PINO-CCR)—for the traditional simulator. The PINO component enforces fluid-flow physics, whereas the CCR framework enhances prediction accuracy in complex and heterogeneous reservoirs; the results are illustrated in Figure 22. Coupling this proxy with an ensemble smoother accelerated history matching by more than four orders of magnitude while maintaining high accuracy. Importantly, the physics constraints narrow the admissible solution space, eliminating parameter sets that fit the data yet violate flow physics and thereby yielding more stable, trustworthy inversions [146].

Thus, PINN–based assisted history matching no longer seeks any parameter set that matches data, but rather the set that simultaneously honours data and the physics of subsurface flow, laying the foundation for more accurate and reliable reservoir models.

4.6 | Production Optimisation

The ultimate purpose of a history-matched reservoir model is to support forward production forecasting and development-scheme

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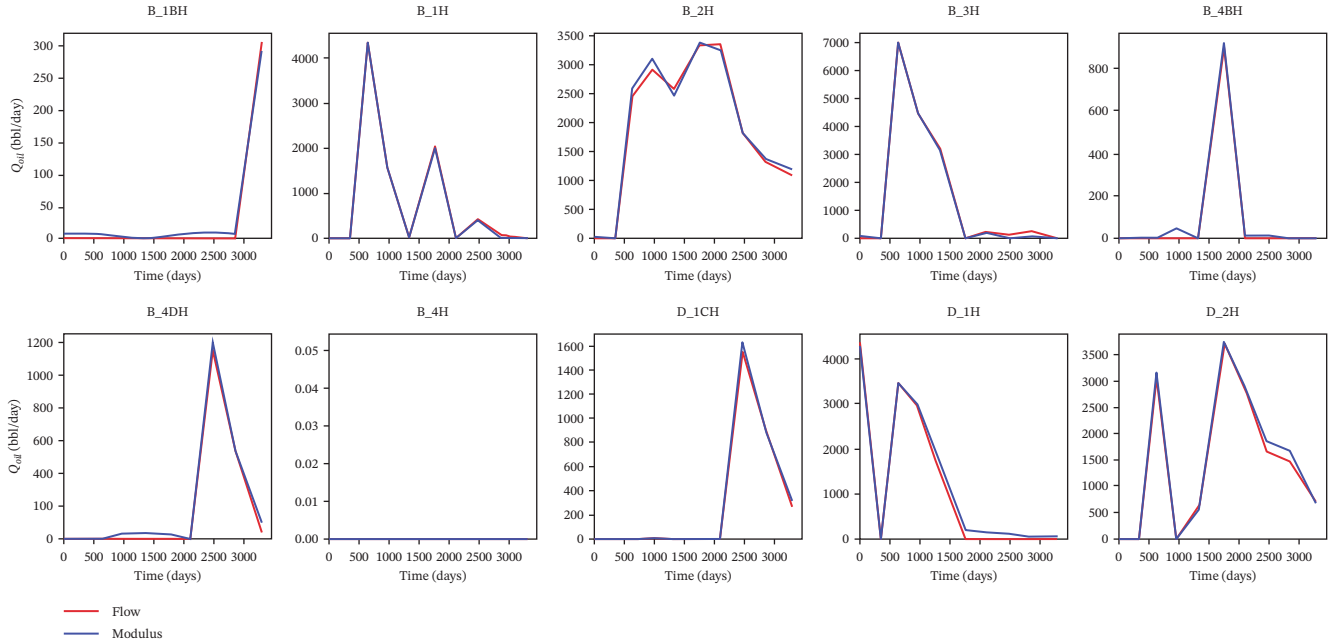

FIGURE 22 | Forwarding of the Norne field comparison between using the true model and PINOCCR model, outputs from peaceman machine for oil production rate [146].

optimisation—both of which hinge on rigorous uncertainty quantification (UQ). Because of subsurface ambiguities, a single deterministic forecast has little value; instead, probabilistic estimates are required to assess the risk-reward profile of alternative operational strategies.

The standard UQ workflow employs Monte Carlo simulation: hundreds or thousands of parameter realisations are drawn from the posterior distribution obtained during history matching, each realisation is propagated through a forward simulator and the ensemble of responses is statistically analysed [147]. Such a brute-force approach is computationally prohibitive with conventional full-physics simulators [148]; hence, rapid machine learning proxies are indispensable [149]. The advantage of a PINN proxy is that every generated scenario not only runs orders of magnitude faster but also remains within the realm of physical plausibility, because the governing flow equations are embedded in the model; this physical consistency renders the resulting UQ more trustworthy.

Production optimisation seeks an optimal set of control variables—new-well locations, injector-producer schedules, fracturing designs and so forth—under operational constraints (surface-facility limits, bottom-hole-pressure bounds, etc.) that maximises a chosen economic objective. The problem is typically solved with heuristic search algorithms such as GAs or PSO.

Khan et al. [150] proposed a hybrid physics-informed machine learning workflow in which reservoir and fracture physics are embedded within a deep neural network 'backbone'. The network is trained on 10,000 synthetic scenarios that fuse physics-based features (rock/fluid properties and fracture monitoring data) with data-driven outputs. Once trained, the PINN proxy predicts key metrics—fluid efficiency, proppant mass and the dimensionless productivity index—at negligible cost. A simulated-annealing algorithm followed by PSO then searches for the fracturing-parameter combination that maximises the productivity index, using the PINN as a rapid evaluator. Because physics constraints are retained, the optimiser explores only feasible design regions. As illustrated in Figure 23, the optimised treatment yields a substantial post-stimulation production uplift [150].

Li and Ma [151] developed a dual-model cooperative-optimisation workflow to realise closed-loop control of well production. The approach first trains two machine learning models: PC–BiGRU, a physics-constrained BiGRU that delivers long-term baseline forecasts, and BiGRU–MTL, a multi-step predictor whose outputs depend on external control variables—specifically choke size and shut-in duration. Using a 15-day sliding window, the short-term forecasts from BiGRU–MTL are continuously compared against the PC–BiGRU baseline. Whenever the predicted production falls below the baseline, an optimisation module is triggered. Choke size and shut-in time are treated as decision variables, and the objective is to maximise the cumulative oil produced within the window. An initial strategy set is generated via Latin hypercube sampling, after which a PSO algorithm searches the decision space. The optimal parameters are fed back to field operations, completing a closed 'monitor-predict-optimise-execute' loop. In a real-field deployment (M field), the scheme increased single-well cumulative oil production by 15.96% over 463 days.

These and related studies demonstrate that embedding diverse physical priors—ranging from conservation laws to empirical formulas—together with advanced optimisation strategies is making PINN–based frameworks increasingly efficient and practical in petroleum engineering. Compared with purely data-driven models, PINNs remain reliable under sparse-data conditions and avoid predictions that violate physical principles. On average, physics-constrained models surpass their data-only counterparts in both predictive accuracy and long-term extrapolative capability [152, 153]. By weaving physics into the network, PINNs alleviate the limitations of 'black box' neural nets: they enforce basic physical consistency and improve robustness when extrapolating beyond the training domain.

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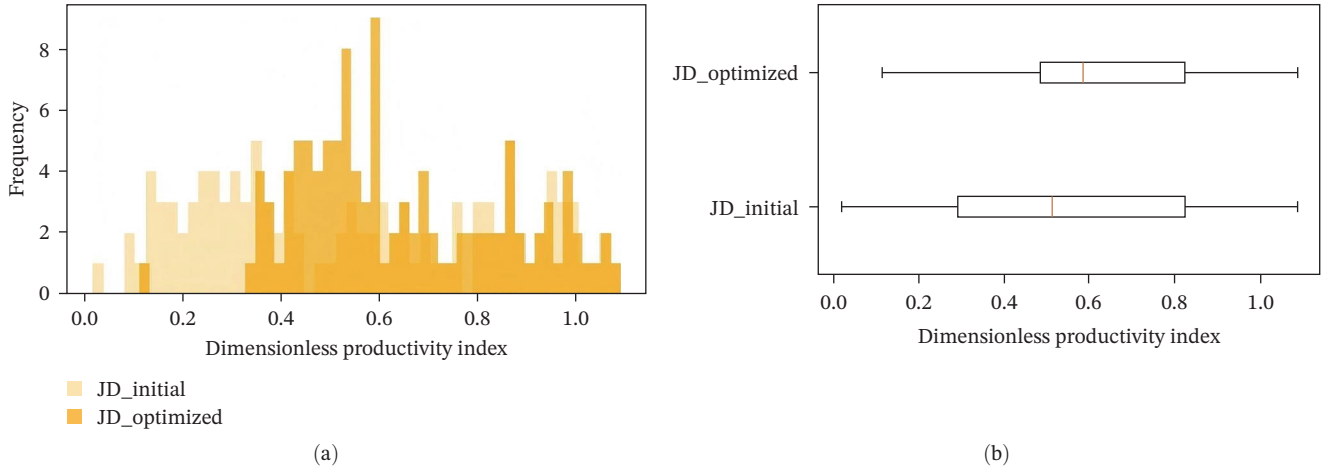

FIGURE 23 | Optimisation results for JD: (a) density plot for initial and optimised values shows the gaussian shift towards right in the increasing JD direction and (b) box plot with the JD spread shows that for lower values an enhancement of upto ~70% can be achieved through optimisation [150].

Admittedly, current PINN implementations can be time-consuming to train and may struggle to converge for highly complex flow phenomena. Nevertheless, recent algorithmic advances and increased computational power are broadening their applicability; the field is moving from proof-of-concept studies toward routine workflows that complement traditional methods. Extensive laboratory and field validations confirm the feasibility of physics-data fusion for production forecasting, establishing a solid foundation for next-generation reservoir-management tools.

5 | Challenges and Future Outlook

Although machine learning methods have made notable strides in forecasting well productivity, they still face severe challenges with respect to model generalisation, physical consistency and data dependency. These hurdles expose current technical limitations while simultaneously charting the trajectory of future research, driving the discipline toward solutions that are more reliable, intelligent and practically valuable.

5.1 | Enhancing Model Generalisation and Data Quality

Machine learning models typically exhibit the well-known dilemma of 'accurate interpolation yet poor extrapolation'. When training data are limited or operating conditions change—such as the commissioning of new wells or the implementation of production measures—their ability to generalise and adapt deteriorates markedly. Compounding this problem, production datasets in most oilfields are sparse, noisy and often drawn from a single distribution, which severely restricts model training. Consequently, future progress must focus on improving both model adaptability and data quality.

5.1.1 | Strengthening Data Quality is a Foundational Prerequisite

5.1.1.1 | Multi-Source Data Fusion A concerted effort is required to integrate heterogeneous information—seismic attributes, well-log signatures, hydraulic-fracturing parameters, production time series and more—into a unified reservoir description. This demands models capable of handling heterogeneous inputs, potentially through hybrid architectures that combine convolutional and graph neural networks. Large foundational models such as Chat-GPT and DeepSeek have recently been explored for integrating disparate oilfield data streams and merit further investigation [154, 155].

5.1.1.2 | Data Augmentation To address data scarcity, augmentation techniques are indispensable. Besides generating synthetic well-log curves via geostatistical simulation [58], generative adversarial networks (GANs) [156] can be employed to create realistic production and logging data that enrich training corpora.

5.1.1.3 | Automated Data Cleansing High-quality preprocessing remains critical. Morozov et al. [24], for example, used collaborative filtering to impute missing values and t-SNE for outlier removal, steps that greatly improved the training dataset.

5.1.2 | Improving Model Adaptability is Equally Crucial

5.1.2.1 | Transfer Learning A powerful remedy is to pre-train general models on data-rich regions—or on high-fidelity simulation data—and then transfer them to data-poor areas [157]. Carpenter's [158] DL framework, for instance, achieved effective cross-field prediction by 'physics-enhancing' its training with synthetic simulation data.

5.1.2.2 | Public Benchmarks and Pre-Trained Foundation Models Establishing large, openly accessible production datasets and benchmark tasks—and developing a 'foundation model' for oil and gas production—would substantially broaden the applicability and generalisation capacity of machine learning tools in data-sparse reservoirs.

By simultaneously elevating data quality and model robustness, the next generation of physically consistent and data-efficient learning frameworks will be better equipped to meet the demanding requirements of modern reservoir management.

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5.2 | Implementation and Balancing of Physical Constraints

The 'black box' character of purely data-driven models remains a principal obstacle to their acceptance in engineering decision making. Such models can generate predictions that violate physical intuition—for example, negative production rates—and their opaque decision pathways hinder practitioner trust. Consequently, the deep integration of physical mechanisms with data-driven learning has become both a consensus and an inevitability in the field.

Physics-constrained machine learning is now one of the most active research frontiers. The paradigm spans (i) hard-constraint approaches typified by PINNs, which embed PDE residuals directly into the loss function to enforce strict adherence to governing laws [123], and (ii) broader soft-constraint or physics-guided methods that blend empirical formulas—such as DCA—or physically meaningful features (permeability, pressure gradients, etc.) into conventional machine learning models. Both routes have proven effective in enhancing physical plausibility and extrapolative power. However, embedding physics rigorously—especially through loss-function enforcement as in PINNs—remains technically challenging.

The training of PINNs is notoriously difficult and often unstable, chiefly owing to the need to balance multiple terms in a composite loss function [159]. A typical formulation may include a data-misfit term, one or more PDE residuals, boundary-condition penalties and initial-condition penalties. These constituents can differ by orders of magnitude, and their gradients may evolve at markedly different rates during optimisation. Without appropriate weighting, a dominant term can 'hijack' the training, leading to divergence or convergence to a poor local optimum [160]. Designing dynamic, self-adaptive schemes for loss-weight tuning is therefore an active but unresolved area of investigation.

A further challenge lies in accurately representing and solving multi-scale and multi-physics couplings within the PINN framework [161]. In unconventional reservoirs, for instance, fluid flow is tightly coupled with rock stress–strain behaviour, complex phase transitions and geochemical reactions. Expressing these highly non-linear and multi-scale phenomena in a form amenable to machine learning integration demands cross-disciplinary expertise and sophisticated modelling skills. Future work should prioritise the development of standardised libraries of physical priors and constraint templates, thereby lowering the barriers to seamless fusion between domain physics and machine learning architectures.

5.3 | Model Interpretability and Causal Inference

As PINN–based systems assume an increasingly pivotal role in production decision-making, mere predictive accuracy is no longer sufficient; it has become essential to understand why a model issues a particular forecast. Consequently, explainable artificial intelligence (XAI) constitutes another critical research frontier for PINNs.

Two principal pathways are currently pursued.

5.3.1 | Post Hoc Interpretability

General XAI techniques—such as SHAP and LIME [162, 163]—are applied to trained networks to quantify the contribution of individual input features (geological and operational parameters) to the forecast [164]. This practice not only fosters engineer trust by verifying whether the decision logic accords with domain expertise, but may also uncover previously unknown physical linkages, enabling a transition from AI–assisted decision making to AI–driven scientific discovery. For instance, Li et al. [165] employed SHAP to analyse a BiLSTM model and quantified the influence of each factor on production; the results corroborated engineering intuition and thereby enhanced model credibility.

5.3.2 | Integrating Causal Inference With Machine Learning

Field data abound with correlations, yet correlation does not imply causation—high-output wells often feature extensive fracturing, but it is unclear whether fracturing causes high production or whether such wells reside in naturally favourable zones. Causal inference seeks to disentangle confounders and identify genuine cause-and-effect relationships. Embedding causal frameworks within machine learning models would more accurately assess the true impact of engineering interventions (fracturing, water flooding, etc.), leading to more scientifically grounded decisions. Min et al. [166], for example, proposed a method that synergises causal inference with machine learning to forecast post-fracture coal-bed-methane well productivity; the technique improved predictive accuracy by 20 % relative to conventional models while yielding causality findings consistent with physical mechanisms, thereby boosting both trust and utility.

Integrating causal inference into machine learning workflows remains technically demanding. Developing accurate and computationally efficient schemes to fuse causal reasoning with data-driven models is poised to become a focal point of future research.

5.4 | System-Level Forecasting and Global Intelligent Optimisation

Current research is shifting from single-well prediction to well-network, system-level forecasting that explicitly accounts for inter-well interference—an intricate yet crucial phenomenon in reservoir development that traditional single-well models cannot capture. Cutting-edge efforts employ graph neural network (GCN) architectures to model well-network topologies and injector-producer relationships explicitly, thereby addressing this challenge.

When prediction is coupled with decision making, an efficient PINN proxy, advanced optimisation algorithms and real-time data streams together constitute the framework of CLRM [167]. In this paradigm, new production data are continuously acquired, the PINN proxy and history match are updated in real time, and the revised model rapidly re-optimises future development strategies. The optimised decisions are then fed back to field operations, forming a persistent 'sense-analyse-decide-execute' cycle [168]. Such a loop has been virtually unattainable with conventional numerical-simulation workflows owing to prohibitive computational cost; it epitomises the disruptive potential of physics-constrained machine learning.

An illustrative example is Machado et al.'s [169] intelligent proxy approach, which endeavours to replace traditional simulators with AI models for injection-production optimisation within a

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closed-loop framework. Although still exploratory, this line of research clearly delineates the trajectory toward fully intelligent field management in future oil- and gas-reservoir operations.

6 | Conclusions

Production-forecasting technology for oil and gas wells is moving decisively beyond empirical heuristics and into the era of data-driven intelligence. This review outlines a clear three-stage progression in the field: an initial phase of data-driven learning, a subsequent embrace of DL architectures, and the latest advance toward physics-informed intelligence.

6.1 | Traditional Machine Learning Laid the Groundwork

Shallow models—exemplified by SVR, RFs and gradient-boosted trees (XGBoost)—provided practical baselines for early single-well forecasting thanks to their efficient non-linear-fitting capability.

6.2 | DL Architectures Delivered the Critical Breakthrough for Sequence Modelling

Gate mechanisms in LSTM and GRU networks mitigated the gradient-vanishing problem of vanilla RNNs, enabling the capture of cyclic fluctuations and shut-in effects in well production. Self-attention-augmented SA–LSTM preserved high-accuracy tracking of decline trends in tight-oil wells even under severe noise. CNN–LSTM hybrids fused spatial features extracted by convolutional layers with temporal dependencies learned by recurrent layers, further elevating predictive accuracy. The advent of emerging models—transformer, GCNs and KANs—has opened entirely new avenues for tackling complex multi-well systems.

6.3 | Physics-Constrained Machine Learning Initiated a Paradigm Shift Toward Physical Consistency

PINNs embed Darcy flow equations and mass-conservation laws directly into the loss function, compelling network outputs to honour subsurface flow physics. The PINN–DD framework, for example, achieved high-fidelity field-wide pressure simulation using only wellhead rates and sparse bottom-hole pressures.

This progression marks more than an algorithmic advance; it reflects a methodological leap—from locally calibrated, feature-engineered models to globally coherent, spatio-temporal frameworks, and ultimately to the deep fusion of physical mechanisms with data intelligence. Such an evolution lays a robust technical foundation for digital CLRM, extending from single-well forecasting to well-cluster optimisation and paving the way toward fully autonomous and intelligent oilfields.

Author Contributions

Shitan Yin: investigation, writing – original draft, writing – review and editing. **Erlong Yang:** supervision, writing – review and editing. **Xianjun Wang:** supervision, writing – review and editing. **Chi Dong:** data validation, manuscript proofreading.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All the data are contained within the article.

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